

## Article

# A Multi-Strategy Collaborative Grey Wolf Optimization Algorithm for UAV Path Planning

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**Abstract:** The Grey Wolf Optimization Algorithm (GWO) is a member of the swarm intelligence algorithm family, which possesses the highlights of easy realization, simple parameter settings and wide applicability. However, in some large-scale application problems, the grey wolf optimization algorithm easily gets trapped in local optima, exhibits poor global exploration ability and suffers from premature convergence. Since grey wolf's update is guided only by the best three wolves, it leads to low population multiplicity and poor global exploration capacity. In response to the above issues, we design a multi-strategy collaborative grey wolf optimization algorithm (NOGWO). Firstly, we use a random walk strategy to extend the exploration scope and enhance the algorithm's global exploration capacity. Secondly, we add an opposition-based learning model influenced by refraction principle to generate an opposite solution for each population, thereby improving population multiplicity and preventing the algorithm from being attracted to local optima. Finally, to balance local exploration and global exploration and elevate the convergence effect, we introduce a novel convergent factor. We conduct experimental testing on NOGWO by using 30 CEC2017 test functions. The experimental outcomes indicate that compared with GWO and some swarm intelligence algorithms, NOGWO has better global exploration capacity and convergence accuracy. In addition, we also apply NOGWO to three engineering problems and an unmanned aerial vehicle path planning problem. The outcomes of the experiment suggest that NOGWO performs well in solving these practical problems.

**Keywords:** grey wolf optimization algorithm; opposition-based learning; refraction principle; random walk; unmanned aerial vehicle path planning



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## 1. Introduction

As a research hotspot in the intelligent optimization field, swarm intelligence algorithms calculate based on the characteristics of natural group behavior. It solves problems by simulating group cooperation and competition. The highlights of swarm intelligence algorithms, such as better adaptability and wider applicability, make them suitable for solving many optimization problems.

At present, there are many types of swarm intelligence algorithms. For example, Particle Swarm Optimization (PSO) [1], which is built upon simulating bird social activities, Genetic Algorithm (GA) [2], which is built upon simulating biological genetics and evolution in natural environments, Ant Colony Optimization (ACO) [3] which is built upon ant's foraging activities, Artificial Bee Colony algorithm (ABC) [4] which derives from bee's foraging activities, etc. They all have been extensively utilized in many fields, including unmanned aerial vehicle path planning [5], job shop scheduling tasks [6], etc.

Grey Wolf Optimizer (GWO) [7] was invented in 2014. It was designed by Mirjalili et al., and its inspiration stems from the cooperative activities of grey wolf populations in the natural world. GWO has been widely welcomed by researchers in recent years because of its rapid convergence speed, simple parameter settings and strong stability. It has developed rapidly in both theoretical research and practical applications. For example,

Rajakumar et al. [8] designed Accelerated Grey Wolf Optimization (AGWO) by introducing mathematical models, diversity measures and parameter adjustment strategies to increase local and global exploration capacities and applied it to solve global optimization problems. Mahdis et al. [9] designed a Grey Wolf Optimizer based on representativeness (R-GWO) to address the issue of low population multiplicity in GWO and utilized it for solving engineering problems. Guha et al. [10] proposed a quasi-opposing grey wolf optimization algorithm (QOGWO) by combining opposition-based learning with a grey wolf optimization algorithm to solve power system control problems. Yang et al. [11] designed the enhanced Grey Wolf Optimization Algorithm (IGWO) by adopting a new convergence factor strategy and utilized it to tackle mechanical design problems. Ou et al. [12] combined the advantages of clone selection algorithms, such as mutation and cloning strategies, with the grey wolf algorithm to strengthen its capacity of jumping out of local optima. They then designed a hybrid grey wolf optimization algorithm (pGWO-CSA) and successfully utilized it to tackle robot path planning problems.

Nonetheless, GWO has some shortcomings, including weak global exploration ability, easy settling into local optima and low optimization precision. It is attributable to GWO only utilizes the best three wolves to lead the population during population updates, resulting in low population multiplicity. Once the three wolves with the best position fall into a local optimal solution, it is challenging for them to leap out and ultimately miss the global optimal. To address these issues, this paper designs a multi-strategy collaborative grey wolf optimization algorithm (NOGWO) that combines an opposition-based learning [13] model strategy influenced by refraction principle [14] and a random walk strategy. By introducing an opposition-based learning model influenced by refraction principle, the population multiplicity is improved and the algorithm can avoid being attracted to local optima. By introducing a random walk strategy, the exploration scope is broadened and the global exploration capability is enhanced. In addition, to improve the algorithm's convergence speed and precision as well as balance local and global exploration, this paper also introduces a nonlinear convergence parameter. Therefore, NOGWO consistently avoids getting trapped in local optima, even during the later stage of iteration, thus facilitating the exploration for global optima and improving convergence accuracy.

The paper's structure outlines as below. Section 2 presents the original grey wolf algorithm and its implementation steps. In Section 3, a detailed introduction is given to the three strategies adopted by NOGWO and its algorithm structure. In Section 4, an experimental comparative analysis is conducted on NOGWO, GWO and some swarm intelligence algorithms. Section 5 discusses the application of NOGWO in unmanned aerial vehicle path planning problem. Section 6 discusses and analyzes the highlights and drawbacks of NOGWO. Finally, the conclusion is drawn in Section 7.

## 2. Grey Wolf Optimization Algorithm

The inspiration for grey wolf algorithm comes from the behavior of grey wolf populations in nature. Its author, Mirjalili et al. [7], designed grey wolf algorithm by observing the social behavior and chasing strategies of grey wolf groups and applied it to the design of optimization algorithms. Grey wolves are a highly hierarchical group of animals that increase the success rate of hunting through cooperation and coordination. A grey wolf pack typically has one or several leaders who are responsible for leading the pack in searching for prey, capturing prey and distributing food. Other members participate in the hunting process by chasing prey and collaborating on capture. In GWO, grey wolf population has four levels:  $\omega$  wolf,  $\delta$  wolf,  $\beta$  wolf and  $\alpha$  wolf. Amidst them,  $\alpha$  wolf has the highest social level,  $\beta$  wolf has the second highest social level,  $\delta$  wolf has the third highest social level,  $\omega$  wolf has the lowest social level. GWO divides the collective activities of grey wolves into three actions: surrounding prey, attacking prey and questing for prey. The relevant mathematical models are stated as below.

### 2.1. Surrounding Prey

Grey wolves are group hunting animals. When they discover prey, they first surround it for better hunting. The mathematical expressions of encircling prey are described in Equations (1) and (2).

$$D = |C \times X_{prey}(k) - X(k)| \quad (1)$$

$$X(k+1) = X_{prey}(k) - A \times D \quad (2)$$

where  $k$  is the present iteration count. Equation (1) is utilized to calculate the length between individual grey wolves and prey,  $D$  represents the length between grey wolves and prey and  $X(k)$  is grey wolf's location vector in the current iteration. Equation (2) is utilized to calculate the next position of individual grey wolves,  $X(k+1)$  is grey wolf's location vector in the next iteration.  $X_{prey}(k)$  typifies the location vector of the prey in the current iteration.  $A$  and  $C$  are two factor vectors, and their calculation formulas are shown in Equations (3)–(5).

$$a = 2 \times (1 - \frac{k}{K}) \quad (3)$$

$$A = 2 \times a \times rand_1 - a \quad (4)$$

$$C = 2 \times rand_2 \quad (5)$$

where  $rand_1$  and  $rand_2$  are uncertain vectors between 0 and 1,  $a$  is a linear convergent factor which descends from 2 to 0 in a straight line.  $K$  is the max iteration count and  $k$  represents the current iteration count. In this paper, the role of  $K$  is to determine the value of the parameter  $a$ , as shown in Equation (3). Additionally,  $K$  is also the standard for the algorithm to stop iteration. To make a fair comparison with the original algorithm, the value of  $K$  is consistent with the value of max iteration count in the original grey wolf algorithm.

In surrounding prey,  $A$  is a pivotal vector. If  $|A| > 1$ , grey wolf will be away from its prey, it expands beyond its encirclement and conducts a larger and wider exploration, which is beneficial for performing an overall exploration. If  $|A| \leq 1$ , grey wolf will approach the prey and shrinks into the encirclement, which is helpful for regional exploration.

### 2.2. Attacking Prey

In GWO, wolf pack has four levels:  $\omega$  wolf,  $\delta$  wolf,  $\beta$  wolf and  $\alpha$  wolf. Amidst them, the wolves with the best three locations are  $\alpha$  wolf,  $\beta$  wolf and  $\delta$  wolf. The remaining wolves only update their locations directed by the locations of  $\alpha$  wolf,  $\beta$  wolf and  $\delta$  wolf. The best three wolves lead numerous  $\omega$  wolves to attack their prey. The mathematical model of four wolves attacking prey is shown in Equations (6)–(12).

$$D_\alpha = |C_1 \times X_\alpha(k) - X(k)| \quad (6)$$

$$D_\beta = |C_2 \times X_\beta(k) - X(k)| \quad (7)$$

$$D_\delta = |C_3 \times X_\delta(k) - X(k)| \quad (8)$$

$$X_1 = X_\alpha(k) - A_1 \times D_\alpha \quad (9)$$

$$X_2 = X_\beta(k) - A_2 \times D_\beta \quad (10)$$

$$X_3 = X_\delta(k) - A_3 \times D_\delta \quad (11)$$

$$X(k+1) = \frac{(X_1 + X_2 + X_3)}{3} \quad (12)$$

where  $X_\alpha(k)$ ,  $X_\beta(k)$ ,  $X_\delta(k)$  and  $X(k)$  typify the locations of  $\alpha$  wolf,  $\beta$  wolf,  $\delta$  wolf and the current  $\omega$  wolf when iteration count is  $k$ .  $C_1$ ,  $C_2$ ,  $C_3$  are three uncertain vectors computed by Equation (5).  $D_\delta$ ,  $D_\beta$ ,  $D_\alpha$  typify the length between the present  $\omega$  wolf and  $\delta$  wolf,  $\beta$  wolf,  $\alpha$  wolf respectively.  $A_1$ ,  $A_2$ ,  $A_3$  are three factor vectors determined by Equation (4).  $X_1$ ,  $X_2$ ,  $X_3$  are vectors for the current  $\omega$  wolf to obtain new positions under the guidance of  $\beta$  wolf,  $\alpha$  wolf and  $\delta$  wolf respectively. Equation (12) updates the current  $\omega$  wolf's location and determines their final locations.

### 2.3. Questing Prey

In GWO, two coefficient vectors  $A$  and  $C$  are set to enable grey wolf to quest for prey better. The parameter  $A$  is utilized to balance grey wolf's global and local exploration capacities. The parameter  $C$  enhances the randomness during the exploration procedure. According to Equation (5), the value of coefficient vector  $C$  is an uncertain digit between 0 and 2, which can avoid getting stuck in local optima.

## 3. A Multi-Strategy Collaborative Grey Wolf Optimization Algorithm

To settle the issue of getting stuck in local optima and premature convergence in complex optimization problems with large scale of grey wolf optimization algorithm, this paper designs a multi-strategy collaborative grey wolf optimization algorithm (NOGWO). NOGWO has introduced three optimization strategies, namely nonlinear convergence factor strategy, opposition-based learning model strategy based on refraction principle and random walk strategy.

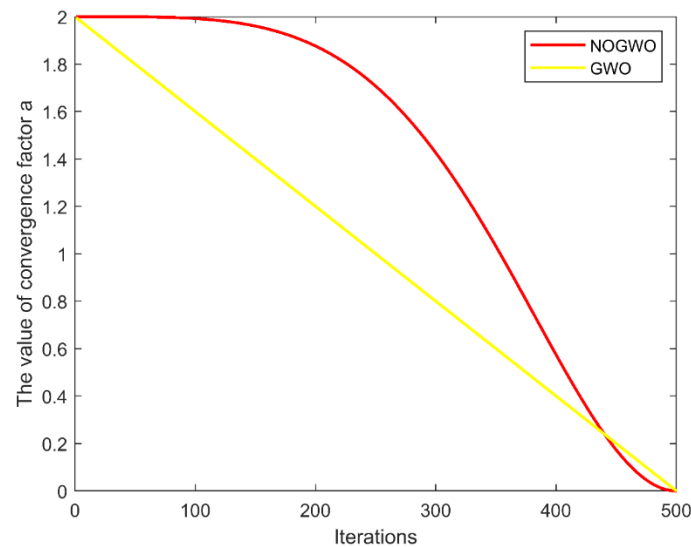
### 3.1. Nonlinear Convergence Factor Strategy

In GWO, local and global exploration of grey wolf is determined by the coefficient vector  $A$ . According to Equation (4), the value of  $A$  is impacted by convergent factor  $a$ . In the case of  $a < 1$ ,  $|A| < 1$ , grey wolf is directed by  $\alpha$  wolf,  $\beta$  wolf,  $\delta$  wolf, approaches the prey and conducts local explorations. In the case of  $a \geq 1$ , the closer the value of  $a$  is to 2, the greater the likelihood of  $|A| \geq 1$  and the easier it is for grey wolf to move away from their prey and spread to a larger range, conducting a global exploration to find better solutions. In traditional grey wolf optimization algorithms, as the iteration count increases, convergent factor  $a$  descends from 2 to 0 in a straight line, resulting in fewer cases of  $|A| \geq 1$ . Grey wolves have limited generations for global exploration and are often prone to settling into local optima. To solve this problem, in NOGWO, a novel convergent factor is adopted. The value of the convergent factor  $a$  is given by the Equation (13).

$$a = 1 + \cos\left(\left(\frac{k}{K}\right)^m \times \Pi\right) \quad (13)$$

where  $m$  is a constant, this paper sets its value to 2, where  $k$  typifies current iteration count,  $K$  is the max iteration count.  $\Pi$  is a constant.

The comparison of convergent factors  $a$  between NOGWO and GWO is shown in Figure 1. The nonlinear convergent factor strategy can let local exploration and global exploration have a good balance effectively, which can broaden the exploration range of grey wolves and elevate the likelihood of detecting the best solution. During the initial iteration period, the nonlinear convergent factor of NOGWO approaches 2 and decreases slowly, which is beneficial for grey wolf to spread to a larger area for global exploration and explore the finest global solution. In iteration's final phase, the value of  $a$  decreases rapidly, which facilitates approaching the searched finest solution and conducting local exploration.



**Figure 1.** Comparison figure of nonlinear and linear convergent factors.

### 3.2. Opposition-Based Learning Model Strategy Influenced by Refraction Principle

Tizhoosh proposed an opposition-based learning strategy in 2005. Opposition-based learning strategy is an extensively utilized and effective swarm intelligence algorithm optimization strategy [15]. Although Opposition-based learning strategy can increase population multiplicity and provide a larger exploration scope, in iteration's final phase, once grey wolves with higher social levels settle into the local finest solution, the surrounding grey wolves will quickly gather around them, ultimately leading to settling into local optima, which will result in poor convergence and optimization effects. To improve the optimization effect of opposition-based learning strategy, Shao P et al. [16] integrated refraction principle model into the opposition-based learning strategy, proposed an opposition-based learning model influenced by refraction principle and applied it to elevate the capacity of particle swarm optimization algorithm. But we have not seen an algorithm that combines the opposition-based learning model strategy influenced by refraction principle with grey wolf algorithm. The mathematical model of the strategy is as stated in Equation (14).

$$X_{newopp} = \frac{(ub + lb)}{2} + \frac{(ub + lb)}{2 \times a \times b} - \frac{X}{a \times b} \quad (14)$$

where  $X$  is existing solution,  $a$  and  $b$  are two constant parameters.  $lb$  and  $ub$  represent the bottom and top limits of exploration scope.  $X_{newopp}$  typifies the solution generated by an opposition-based learning model influenced by refraction principle. Compared with the opposition-based learning strategy, the biggest feature of the opposition-based learning model influenced by refraction principle is the introduction of two constant parameters,  $a$  and  $b$ , where  $a$  called scaling factor and  $b$  called fine-tuning factor, both utilized to adjust the distance between the existing solution and the finest solution, so that the generated solution is closer to the optima. The two parameters are elaborately described in reference [16].

### 3.3. Random Walk Strategy

The exploration process of grey wolves is often not smooth, sometimes they may fall into some obstacles, making it impossible for grey wolf population to detect prey in the end. In GWO, according to Equation (5), parameter  $C$  is an uncertain digit between 0 and 2. The parameter  $C$  increases the randomness in GWO's exploration for the global optima, making the position update more randomly, which is conducive to jumping out of local finest solution. To further enhance the randomness of grey wolf exploration process and increase

the likelihood of exploring unsearched areas, a random walk strategy is introduced, it is stated as Equation (15).

$$X(k+1) = \begin{cases} X_r(k) - r_1 \times |X_r(k) - 2 \times r_2 \times X(k)| & r_3 \geq 0.5 \\ X_\alpha(k) - X_{avg}(k) - r_4 \times (r_5 \times (ub - lb) + lb) & r_3 < 0.5 \end{cases} \quad (15)$$

where  $k$  is iteration count,  $r_5, r_4, r_3, r_2, r_1$ , are all uncertain digits between 0 and 1.  $X_r(k)$ ,  $X_{avg}(k)$ ,  $X_\alpha(k)$  represent the position of any grey wolf, the average position of all grey wolves, the position of  $\alpha$  wolf when iteration count is  $k$ , respectively.  $lb$  and  $ub$  represent the bottom and top limits of exploration scope.  $X(k)$  and  $X(k+1)$  represent the current location of grey wolf and the location of grey wolves in next generation respectively. When the position obtained by the random walk strategy is better than that obtained by Equation (12), apply the random walk strategy to update grey wolf's position. Otherwise, apply Equation (12) to update grey wolf's position.

### 3.4. Time Complexity Analysis

For NOGWO, most of its time is spent comparing the current solution with its opposite solution  $X_{newopp}$  created by Equation (14) or new solution created by Equation (15) and updating the population. Suppose the population size is  $N$  and the dimension is  $D$ . For GWO, the time of population location update is  $ND$  and the number of population comparison is  $N$ . Therefore, the total time of GWO is  $ND+N$  and the time complexity is  $O(ND)$ . For NOGWO, due to the addition of random walk strategy and opposition-based learning model strategy influenced by refraction principle, the number of population comparison is  $3N$  and the time of population location update is  $ND$ . Thus, the total time of NOGWO is  $3N+ND$  and the time complexity is  $O(ND)$ .

The algorithm flowcharts of GWO and NOGWO are shown in Figure 2, where  $k$  represents iteration count,  $K$  is the max iteration count,  $N$  represents the count of grey wolf populations.

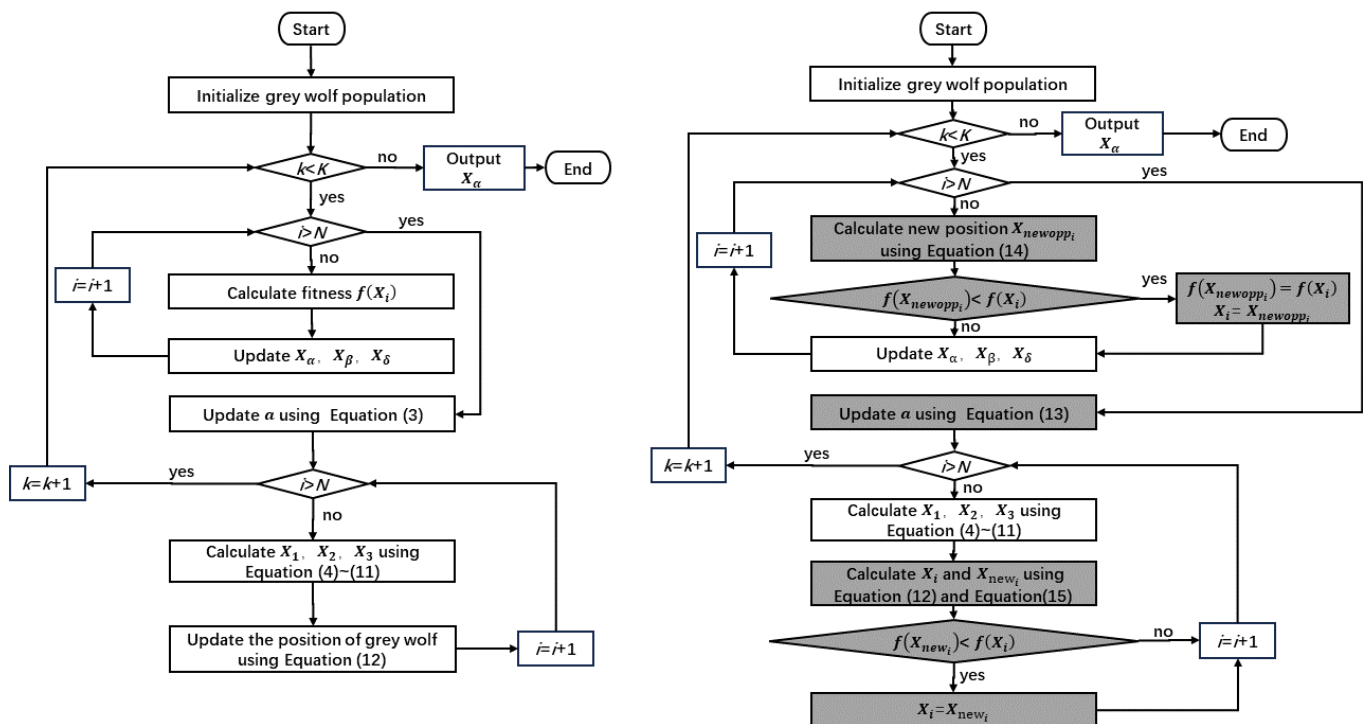


Figure 2. Algorithm flowcharts for GWO (left) and NOGWO (right).



## 4. Experimental and Analysis

### 4.1. Test Functions and Parameter Settings

To fully test the algorithm performance of NOGWO and conduct feasibility analysis on the opposition-based learning model strategy influenced by refraction principle and the random walk strategy, 30 CEC2017 [17] test functions were selected for testing. The function set is categorized into composition (F21–F30), hybrid (F11–F20), simple multimodal (F4–F10) and unimodal functions (F1–F3). In unimodal functions, only a solitary optimum exists, while simple multimodal functions manifest multiple peaks within their construction. Composition functions create complex landscapes by sequentially combining simple functions, while hybrid functions generate diverse search spaces with both local and global optima by mixing different types of functions. They all present challenges to optimization algorithms. The indicators chosen to evaluate the algorithm's performance are maximum, standard deviation, minimum and mean. For the reason the mean is more sensitive to outliers and can respond more acutely to extreme situations, while the median is less affected by outliers, the mean rather than the median is chosen as an indicator. In this experiment, grey wolf population's scale is 30, the iteration count is 500, every algorithm runs 30 times on every test function. In accordance with the method for determining the two parameters of opposition-based learning model strategy influenced by refraction principle in reference [16], in NOGWO, the parameters  $a$  and  $b$  mentioned in Equation (14) are set to 0.48 and 2 respectively.

### 4.2. Comparison with GWO and Its Variants

To test NOGWO's competence and conduct feasibility analysis on the opposition-based learning model strategy influenced by refraction principle and the random walk strategy, this paper selects grey wolf algorithm OGWO based on traditional opposition-based learning models and grey wolf algorithm newGWO using an opposition-based learning model strategy influenced by refraction principle for comparison. Set dimension to 30 and 50 respectively and run the four swarm intelligence algorithms 30 times on 30 CEC2017 test functions. Record the finest fitness value of every run, compare and analyze their maximums, mean values, minimums and standard deviation values. Table 1 presents the experimental outcomes. In addition, if NOGWO achieves the best outcome on the test function, it should be marked in bold.

**Table 1.** Dim = 30.

| Function | Algorithm | Mean                                    | Std                                     | Max                                     | Min                                  |
|----------|-----------|-----------------------------------------|-----------------------------------------|-----------------------------------------|--------------------------------------|
| F1       | GWO       | $2.74 \times 10^9$                      | $2.03 \times 10^9$                      | $9.05 \times 10^9$                      | $2.17 \times 10^8$                   |
|          | OGWO      | $2.84 \times 10^9$                      | $1.73 \times 10^9$                      | $7.46 \times 10^9$                      | $1.83 \times 10^8$                   |
|          | newOGWO   | $2.71 \times 10^9$                      | $1.71 \times 10^9$                      | $6.98 \times 10^9$                      | $1.89 \times 10^8$                   |
|          | NOGWO     | $3.67 \times 10^9$                      | $2.24 \times 10^9$                      | $1.05 \times 10^{10}$                   | $9.13 \times 10^8$                   |
| F2       | GWO       | $7.78 \times 10^{32}$                   | $2.42 \times 10^{33}$                   | $1.21 \times 10^{34}$                   | $1.87 \times 10^{19}$                |
|          | OGWO      | $4.52 \times 10^{33}$                   | $2.50 \times 10^{34}$                   | $1.39 \times 10^{35}$                   | $6.39 \times 10^{18}$                |
|          | newOGWO   | $1.25 \times 10^{32}$                   | $6.24 \times 10^{32}$                   | $3.48 \times 10^{33}$                   | $6.81 \times 10^{18}$                |
|          | NOGWO     | <b><math>4.53 \times 10^{31}</math></b> | <b><math>1.96 \times 10^{32}</math></b> | <b><math>1.08 \times 10^{33}</math></b> | $2.12 \times 10^{20}$                |
| F3       | GWO       | $6.31 \times 10^4$                      | $9.66 \times 10^3$                      | $8.57 \times 10^4$                      | $3.71 \times 10^4$                   |
|          | OGWO      | $5.74 \times 10^4$                      | $6.91 \times 10^3$                      | $6.99 \times 10^4$                      | $4.16 \times 10^4$                   |
|          | newOGWO   | $5.24 \times 10^4$                      | $8.56 \times 10^3$                      | $6.75 \times 10^4$                      | $3.24 \times 10^4$                   |
|          | NOGWO     | <b><math>4.54 \times 10^4</math></b>    | $9.29 \times 10^3$                      | <b><math>6.06 \times 10^4</math></b>    | <b><math>2.58 \times 10^4</math></b> |
| F4       | GWO       | $6.52 \times 10^2$                      | $1.60 \times 10^2$                      | $1.44 \times 10^3$                      | $5.02 \times 10^2$                   |
|          | OGWO      | $6.65 \times 10^2$                      | $1.15 \times 10^2$                      | $9.54 \times 10^2$                      | $5.22 \times 10^2$                   |
|          | newOGWO   | $6.52 \times 10^2$                      | $1.36 \times 10^2$                      | $1.12 \times 10^3$                      | $5.09 \times 10^2$                   |
|          | NOGWO     | <b><math>6.44 \times 10^2</math></b>    | <b><math>9.60 \times 10^1</math></b>    | <b><math>9.51 \times 10^2</math></b>    | $5.35 \times 10^2$                   |

Table 1. Cont.

| Function | Algorithm | Mean                                 | Std                                  | Max                                  | Min                                  |
|----------|-----------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| F5       | GWO       | $6.26 \times 10^2$                   | $4.30 \times 10^1$                   | $7.33 \times 10^2$                   | $5.63 \times 10^2$                   |
|          | OGWO      | $6.24 \times 10^2$                   | $3.78 \times 10^1$                   | $7.30 \times 10^2$                   | $5.71 \times 10^2$                   |
|          | newOGWO   | $6.29 \times 10^2$                   | $3.87 \times 10^1$                   | $7.44 \times 10^2$                   | $5.72 \times 10^2$                   |
|          | NOGWO     | $6.51 \times 10^2$                   | $4.86 \times 10^1$                   | $7.72 \times 10^2$                   | $5.89 \times 10^2$                   |
| F6       | GWO       | $6.14 \times 10^2$                   | $6.13 \times 10^0$                   | $6.29 \times 10^2$                   | $6.06 \times 10^2$                   |
|          | OGWO      | $6.12 \times 10^2$                   | $4.24 \times 10^0$                   | $6.21 \times 10^2$                   | $6.04 \times 10^2$                   |
|          | newOGWO   | $6.13 \times 10^2$                   | $4.51 \times 10^0$                   | $6.25 \times 10^2$                   | $6.04 \times 10^2$                   |
|          | NOGWO     | <b><math>6.12 \times 10^2</math></b> | $5.58 \times 10^0$                   | $6.27 \times 10^2$                   | <b><math>6.04 \times 10^2</math></b> |
| F7       | GWO       | $9.15 \times 10^2$                   | $6.10 \times 10^1$                   | $1.06 \times 10^3$                   | $8.30 \times 10^2$                   |
|          | OGWO      | $8.99 \times 10^2$                   | $4.01 \times 10^1$                   | $1.00 \times 10^3$                   | $8.44 \times 10^2$                   |
|          | newOGWO   | $8.99 \times 10^2$                   | $4.50 \times 10^1$                   | $1.00 \times 10^3$                   | $8.25 \times 10^2$                   |
|          | NOGWO     | $9.36 \times 10^2$                   | $5.75 \times 10^1$                   | $1.06 \times 10^3$                   | $8.44 \times 10^2$                   |
| F8       | GWO       | $9.11 \times 10^2$                   | $3.05 \times 10^1$                   | $1.02 \times 10^3$                   | $8.81 \times 10^2$                   |
|          | OGWO      | $9.07 \times 10^2$                   | $3.23 \times 10^1$                   | $1.03 \times 10^3$                   | $8.69 \times 10^2$                   |
|          | newOGWO   | $9.07 \times 10^2$                   | $3.04 \times 10^1$                   | $1.04 \times 10^3$                   | $8.68 \times 10^2$                   |
|          | NOGWO     | $9.41 \times 10^2$                   | $5.21 \times 10^1$                   | $1.06 \times 10^3$                   | $8.91 \times 10^2$                   |
| F9       | GWO       | $2.95 \times 10^3$                   | $1.40 \times 10^3$                   | $6.85 \times 10^3$                   | $1.44 \times 10^3$                   |
|          | OGWO      | $2.20 \times 10^3$                   | $7.00 \times 10^2$                   | $4.58 \times 10^3$                   | $1.37 \times 10^3$                   |
|          | newOGWO   | $2.63 \times 10^3$                   | $1.19 \times 10^3$                   | $7.31 \times 10^3$                   | $9.88 \times 10^2$                   |
|          | NOGWO     | $2.83 \times 10^3$                   | $9.26 \times 10^2$                   | $5.09 \times 10^3$                   | <b><math>1.36 \times 10^3</math></b> |
| F10      | GWO       | $5.17 \times 10^3$                   | $1.56 \times 10^3$                   | $9.29 \times 10^3$                   | $3.26 \times 10^3$                   |
|          | OGWO      | $6.45 \times 10^3$                   | $1.82 \times 10^3$                   | $8.96 \times 10^3$                   | $3.93 \times 10^3$                   |
|          | newOGWO   | $6.44 \times 10^3$                   | $2.00 \times 10^3$                   | $9.01 \times 10^3$                   | $3.45 \times 10^3$                   |
|          | NOGWO     | $7.58 \times 10^3$                   | $1.58 \times 10^3$                   | $9.15 \times 10^3$                   | $4.23 \times 10^3$                   |
| F11      | GWO       | $2.63 \times 10^3$                   | $1.26 \times 10^3$                   | $4.89 \times 10^3$                   | $1.33 \times 10^3$                   |
|          | OGWO      | $2.41 \times 10^3$                   | $1.11 \times 10^3$                   | $5.18 \times 10^3$                   | $1.38 \times 10^3$                   |
|          | newOGWO   | $2.16 \times 10^3$                   | $8.58 \times 10^2$                   | $4.35 \times 10^3$                   | $1.29 \times 10^3$                   |
|          | NOGWO     | <b><math>1.42 \times 10^3</math></b> | <b><math>9.19 \times 10^1</math></b> | <b><math>1.60 \times 10^3</math></b> | <b><math>1.29 \times 10^3</math></b> |
| F12      | GWO       | $1.73 \times 10^8$                   | $3.45 \times 10^8$                   | $1.85 \times 10^9$                   | $1.33 \times 10^7$                   |
|          | OGWO      | $1.04 \times 10^8$                   | $9.05 \times 10^7$                   | $3.30 \times 10^8$                   | $1.04 \times 10^7$                   |
|          | newOGWO   | $9.99 \times 10^7$                   | $7.59 \times 10^7$                   | $2.83 \times 10^8$                   | $3.95 \times 10^6$                   |
|          | NOGWO     | $1.18 \times 10^8$                   | $1.31 \times 10^8$                   | $5.75 \times 10^8$                   | $7.17 \times 10^6$                   |
| F13      | GWO       | $1.76 \times 10^7$                   | $3.80 \times 10^7$                   | $1.20 \times 10^8$                   | $5.58 \times 10^4$                   |
|          | OGWO      | $7.88 \times 10^6$                   | $2.11 \times 10^7$                   | $9.60 \times 10^7$                   | $4.84 \times 10^4$                   |
|          | newOGWO   | $4.04 \times 10^7$                   | $1.00 \times 10^8$                   | $3.47 \times 10^8$                   | $7.06 \times 10^4$                   |
|          | NOGWO     | $8.67 \times 10^6$                   | $2.34 \times 10^7$                   | <b><math>9.46 \times 10^7</math></b> | <b><math>2.36 \times 10^4</math></b> |
| F14      | GWO       | $6.63 \times 10^5$                   | $7.02 \times 10^5$                   | $2.15 \times 10^6$                   | $4.76 \times 10^3$                   |
|          | OGWO      | $7.28 \times 10^5$                   | $7.54 \times 10^5$                   | $2.71 \times 10^6$                   | $5.56 \times 10^3$                   |
|          | newOGWO   | $5.73 \times 10^5$                   | $5.11 \times 10^5$                   | $1.79 \times 10^6$                   | $3.11 \times 10^3$                   |
|          | NOGWO     | <b><math>2.49 \times 10^5</math></b> | <b><math>2.97 \times 10^5</math></b> | <b><math>9.62 \times 10^5</math></b> | $3.79 \times 10^3$                   |
| F15      | GWO       | $1.98 \times 10^6$                   | $6.19 \times 10^6$                   | $3.39 \times 10^7$                   | $1.90 \times 10^4$                   |
|          | OGWO      | $8.82 \times 10^5$                   | $1.49 \times 10^6$                   | $5.28 \times 10^6$                   | $3.50 \times 10^4$                   |
|          | newOGWO   | $8.38 \times 10^5$                   | $1.36 \times 10^6$                   | $4.57 \times 10^6$                   | $1.27 \times 10^4$                   |
|          | NOGWO     | <b><math>8.32 \times 10^5</math></b> | <b><math>1.30 \times 10^6</math></b> | $5.67 \times 10^6$                   | $2.64 \times 10^4$                   |
| F16      | GWO       | $2.69 \times 10^3$                   | $3.95 \times 10^2$                   | $3.67 \times 10^3$                   | $1.85 \times 10^3$                   |
|          | OGWO      | $2.60 \times 10^3$                   | $2.37 \times 10^2$                   | $3.09 \times 10^3$                   | $2.16 \times 10^3$                   |
|          | newOGWO   | $2.65 \times 10^3$                   | $4.25 \times 10^2$                   | $3.36 \times 10^3$                   | $1.90 \times 10^3$                   |
|          | NOGWO     | $2.78 \times 10^3$                   | $4.10 \times 10^2$                   | $3.84 \times 10^3$                   | $2.24 \times 10^3$                   |



Table 1. Cont.

| Function | Algorithm | Mean                                 | Std                                  | Max                                  | Min                                  |
|----------|-----------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| F17      | GWO       | $2.04 \times 10^3$                   | $1.64 \times 10^2$                   | $2.40 \times 10^3$                   | $1.82 \times 10^3$                   |
|          | OGWO      | $2.11 \times 10^3$                   | $1.98 \times 10^2$                   | $2.50 \times 10^3$                   | $1.81 \times 10^3$                   |
|          | newOGWO   | $2.03 \times 10^3$                   | $1.66 \times 10^2$                   | $2.38 \times 10^3$                   | $1.80 \times 10^3$                   |
|          | NOGWO     | <b><math>2.02 \times 10^3</math></b> | <b><math>1.41 \times 10^2</math></b> | <b><math>2.36 \times 10^3</math></b> | <b><math>1.79 \times 10^3</math></b> |
| F18      | GWO       | $4.68 \times 10^6$                   | $1.29 \times 10^7$                   | $7.29 \times 10^7$                   | $1.81 \times 10^5$                   |
|          | OGWO      | $1.91 \times 10^6$                   | $1.95 \times 10^6$                   | $7.27 \times 10^6$                   | $1.29 \times 10^5$                   |
|          | newOGWO   | $1.85 \times 10^6$                   | $3.16 \times 10^6$                   | $1.81 \times 10^7$                   | $3.22 \times 10^4$                   |
|          | NOGWO     | <b><math>9.51 \times 10^5</math></b> | <b><math>1.24 \times 10^6</math></b> | <b><math>6.08 \times 10^6</math></b> | $4.69 \times 10^4$                   |
| F19      | GWO       | $2.08 \times 10^6$                   | $6.65 \times 10^6$                   | $3.74 \times 10^7$                   | $1.64 \times 10^4$                   |
|          | OGWO      | $2.00 \times 10^6$                   | $6.99 \times 10^6$                   | $3.95 \times 10^7$                   | $8.14 \times 10^3$                   |
|          | newOGWO   | $1.99 \times 10^6$                   | $6.09 \times 10^6$                   | $3.42 \times 10^7$                   | $2.54 \times 10^4$                   |
|          | NOGWO     | <b><math>1.45 \times 10^6</math></b> | <b><math>2.43 \times 10^6</math></b> | <b><math>1.33 \times 10^7</math></b> | <b><math>7.01 \times 10^3</math></b> |
| F20      | GWO       | $2.52 \times 10^3$                   | $1.86 \times 10^2$                   | $2.90 \times 10^3$                   | $2.18 \times 10^3$                   |
|          | OGWO      | $2.60 \times 10^3$                   | $2.34 \times 10^2$                   | $3.12 \times 10^3$                   | $2.23 \times 10^3$                   |
|          | newOGWO   | $2.48 \times 10^3$                   | $2.31 \times 10^2$                   | $2.98 \times 10^3$                   | $2.20 \times 10^3$                   |
|          | NOGWO     | <b><math>2.43 \times 10^3</math></b> | <b><math>1.58 \times 10^2</math></b> | <b><math>2.89 \times 10^3</math></b> | $2.21 \times 10^3$                   |
| F21      | GWO       | $2.42 \times 10^3$                   | $4.02 \times 10^1$                   | $2.54 \times 10^3$                   | $2.37 \times 10^3$                   |
|          | OGWO      | $2.41 \times 10^3$                   | $4.69 \times 10^1$                   | $2.56 \times 10^3$                   | $2.35 \times 10^3$                   |
|          | newOGWO   | $2.41 \times 10^3$                   | $3.24 \times 10^1$                   | $2.52 \times 10^3$                   | $2.35 \times 10^3$                   |
|          | NOGWO     | $2.44 \times 10^3$                   | $4.90 \times 10^1$                   | $2.54 \times 10^3$                   | $2.37 \times 10^3$                   |
| F22      | GWO       | $5.72 \times 10^3$                   | $2.28 \times 10^3$                   | $1.02 \times 10^4$                   | $2.51 \times 10^3$                   |
|          | OGWO      | $3.09 \times 10^3$                   | $1.32 \times 10^3$                   | $9.61 \times 10^3$                   | $2.39 \times 10^3$                   |
|          | newOGWO   | $2.98 \times 10^3$                   | $5.07 \times 10^2$                   | $4.89 \times 10^3$                   | $2.44 \times 10^3$                   |
|          | NOGWO     | <b><math>2.97 \times 10^3</math></b> | $8.19 \times 10^2$                   | $7.08 \times 10^3$                   | $2.45 \times 10^3$                   |
| F23      | GWO       | $2.79 \times 10^3$                   | $5.37 \times 10^1$                   | $2.92 \times 10^3$                   | $2.73 \times 10^3$                   |
|          | OGWO      | $2.81 \times 10^3$                   | $5.84 \times 10^1$                   | $2.92 \times 10^3$                   | $2.73 \times 10^3$                   |
|          | newOGWO   | $2.79 \times 10^3$                   | $5.35 \times 10^1$                   | $2.92 \times 10^3$                   | $2.71 \times 10^3$                   |
|          | NOGWO     | <b><math>2.78 \times 10^3</math></b> | $5.85 \times 10^1$                   | $2.98 \times 10^3$                   | $2.73 \times 10^3$                   |
| F24      | GWO       | $2.98 \times 10^3$                   | $6.95 \times 10^1$                   | $3.14 \times 10^3$                   | $2.89 \times 10^3$                   |
|          | OGWO      | $2.97 \times 10^3$                   | $7.29 \times 10^1$                   | $3.12 \times 10^3$                   | $2.88 \times 10^3$                   |
|          | newOGWO   | $2.96 \times 10^3$                   | $6.97 \times 10^1$                   | $3.13 \times 10^3$                   | $2.86 \times 10^3$                   |
|          | NOGWO     | $2.98 \times 10^3$                   | <b><math>6.57 \times 10^1</math></b> | <b><math>3.10 \times 10^3</math></b> | $2.88 \times 10^3$                   |
| F25      | GWO       | $3.03 \times 10^3$                   | $4.22 \times 10^1$                   | $3.12 \times 10^3$                   | $2.95 \times 10^3$                   |
|          | OGWO      | $3.02 \times 10^3$                   | $6.58 \times 10^1$                   | $3.28 \times 10^3$                   | $2.93 \times 10^3$                   |
|          | newOGWO   | $3.04 \times 10^3$                   | $1.05 \times 10^2$                   | $3.40 \times 10^3$                   | $2.93 \times 10^3$                   |
|          | NOGWO     | <b><math>3.02 \times 10^3</math></b> | $4.38 \times 10^1$                   | <b><math>3.10 \times 10^3</math></b> | <b><math>2.93 \times 10^3</math></b> |
| F26      | GWO       | $5.07 \times 10^3$                   | $4.22 \times 10^2$                   | $5.93 \times 10^3$                   | $4.46 \times 10^3$                   |
|          | OGWO      | $5.03 \times 10^3$                   | $4.47 \times 10^2$                   | $6.59 \times 10^3$                   | $4.32 \times 10^3$                   |
|          | newOGWO   | $4.92 \times 10^3$                   | $3.90 \times 10^2$                   | $5.79 \times 10^3$                   | $4.33 \times 10^3$                   |
|          | NOGWO     | <b><math>4.88 \times 10^3</math></b> | $6.00 \times 10^2$                   | $6.04 \times 10^3$                   | <b><math>3.65 \times 10^3</math></b> |
| F27      | GWO       | $3.27 \times 10^3$                   | $3.82 \times 10^1$                   | $3.44 \times 10^3$                   | $3.23 \times 10^3$                   |
|          | OGWO      | $3.27 \times 10^3$                   | $3.22 \times 10^1$                   | $3.35 \times 10^3$                   | $3.23 \times 10^3$                   |
|          | newOGWO   | $3.27 \times 10^3$                   | $2.72 \times 10^1$                   | $3.33 \times 10^3$                   | $3.23 \times 10^3$                   |
|          | NOGWO     | <b><math>3.26 \times 10^3</math></b> | $3.01 \times 10^1$                   | $3.36 \times 10^3$                   | <b><math>3.21 \times 10^3</math></b> |
| F28      | GWO       | $3.48 \times 10^3$                   | $1.03 \times 10^2$                   | $3.79 \times 10^3$                   | $3.30 \times 10^3$                   |
|          | OGWO      | $3.48 \times 10^3$                   | $1.53 \times 10^2$                   | $4.18 \times 10^3$                   | $3.36 \times 10^3$                   |
|          | newOGWO   | $3.47 \times 10^3$                   | $9.95 \times 10^1$                   | $3.71 \times 10^3$                   | $3.29 \times 10^3$                   |
|          | NOGWO     | <b><math>3.46 \times 10^3</math></b> | $1.35 \times 10^2$                   | $4.08 \times 10^3$                   | $3.30 \times 10^3$                   |

Table 1. Cont.

| Function              | Algorithm | Mean                                 | Std                | Max                                  | Min                                  |
|-----------------------|-----------|--------------------------------------|--------------------|--------------------------------------|--------------------------------------|
| F29                   | GWO       | $3.97 \times 10^3$                   | $1.90 \times 10^2$ | $4.41 \times 10^3$                   | $3.56 \times 10^3$                   |
|                       | OGWO      | $3.95 \times 10^3$                   | $2.22 \times 10^2$ | $4.52 \times 10^3$                   | $3.56 \times 10^3$                   |
|                       | newOGWO   | $3.91 \times 10^3$                   | $2.01 \times 10^2$ | $4.37 \times 10^3$                   | $3.58 \times 10^3$                   |
|                       | NOGWO     | <b><math>3.86 \times 10^3</math></b> | $2.73 \times 10^2$ | $4.77 \times 10^3$                   | <b><math>3.50 \times 10^3</math></b> |
| F30                   | GWO       | $1.16 \times 10^7$                   | $7.80 \times 10^6$ | $2.81 \times 10^7$                   | $5.77 \times 10^5$                   |
|                       | OGWO      | $1.18 \times 10^7$                   | $7.40 \times 10^6$ | $2.92 \times 10^7$                   | $8.60 \times 10^5$                   |
|                       | newOGWO   | $1.39 \times 10^7$                   | $1.21 \times 10^7$ | $5.36 \times 10^7$                   | $7.73 \times 10^5$                   |
|                       | NOGWO     | <b><math>1.01 \times 10^7</math></b> | $8.11 \times 10^6$ | <b><math>2.81 \times 10^7</math></b> | $6.93 \times 10^5$                   |
| Data analysis results |           | 19/30                                | 10/30              | 13/30                                | 11/30                                |

Table 1 indicates that in the case of 30 dimensions, in terms of average value, NOGWO only lost 11 test functions, which means NOGWO's converging precision is excellent on most test functions. It is worth mentioning that among the ten complex composition functions F21~F30, NOGWO only lost two test functions. Among the ten hybrid functions F11~F20, a total of seven test functions achieved optimal values. It means that when dealing with complex optimization problems, NOGWO often has better optimization ability. In terms of standard deviation, maximum value and minimum value, NOGWO performs slightly better than GWO. In addition, in most test functions such as F11, F19, F22, F26, F28 and F29, the average values obtained by GWO, OGWO, newOGWO and NOGWO decrease sequentially, which fully demonstrates the feasibility of the opposition-based learning model strategy influenced by refraction principle and random walk strategy.

To comprehensively compare NOGWO's competence, set the dimension to 50 and conduct the same experiment again. Table 2 presents the experimental outcomes in 50 dimensions. Table 2 indicates that although the overall performance of NOGWO has declined in high-dimensional situations, in terms of average value, it still achieved the optimal value on nearly half of the test functions. However, in terms of standard deviation, NOGWO's performance is not as outstanding as the average.

Table 2. Dim = 50.

| Function | Algorithm | Mean                                    | Std                                     | Max                                     | Min                                     |
|----------|-----------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| F1       | GWO       | $1.19 \times 10^{10}$                   | $4.29 \times 10^9$                      | $2.06 \times 10^{10}$                   | $4.29 \times 10^9$                      |
|          | OGWO      | $1.09 \times 10^{10}$                   | $4.75 \times 10^9$                      | $1.94 \times 10^{10}$                   | $1.93 \times 10^9$                      |
|          | newOGWO   | $1.04 \times 10^{10}$                   | $3.84 \times 10^9$                      | $1.76 \times 10^{10}$                   | $2.77 \times 10^9$                      |
|          | NOGWO     | $1.54 \times 10^{10}$                   | $5.50 \times 10^9$                      | $2.81 \times 10^{10}$                   | $7.04 \times 10^9$                      |
| F2       | GWO       | $4.74 \times 10^{55}$                   | $2.63 \times 10^{56}$                   | $1.46 \times 10^{57}$                   | $9.55 \times 10^{44}$                   |
|          | OGWO      | $2.58 \times 10^{55}$                   | $1.44 \times 10^{56}$                   | $8.00 \times 10^{56}$                   | $1.02 \times 10^{43}$                   |
|          | newOGWO   | $5.98 \times 10^{59}$                   | $3.33 \times 10^{60}$                   | $1.86 \times 10^{61}$                   | $2.56 \times 10^{45}$                   |
|          | NOGWO     | <b><math>5.62 \times 10^{53}</math></b> | <b><math>3.08 \times 10^{54}</math></b> | <b><math>1.72 \times 10^{55}</math></b> | <b><math>3.34 \times 10^{42}</math></b> |
| F3       | GWO       | $1.68 \times 10^5$                      | $3.11 \times 10^4$                      | $2.38 \times 10^5$                      | $1.17 \times 10^5$                      |
|          | OGWO      | $1.41 \times 10^5$                      | $1.50 \times 10^4$                      | $1.70 \times 10^5$                      | $1.12 \times 10^5$                      |
|          | newOGWO   | $1.47 \times 10^5$                      | $2.16 \times 10^4$                      | $1.92 \times 10^5$                      | $1.11 \times 10^5$                      |
|          | NOGWO     | <b><math>1.26 \times 10^5</math></b>    | $1.65 \times 10^4$                      | <b><math>1.58 \times 10^5</math></b>    | <b><math>9.81 \times 10^4</math></b>    |
| F4       | GWO       | $1.61 \times 10^3$                      | $5.93 \times 10^2$                      | $2.97 \times 10^3$                      | $7.39 \times 10^2$                      |
|          | OGWO      | $1.55 \times 10^3$                      | $6.17 \times 10^2$                      | $2.97 \times 10^3$                      | $7.96 \times 10^2$                      |
|          | newOGWO   | $1.55 \times 10^3$                      | $7.41 \times 10^2$                      | $3.62 \times 10^3$                      | $8.07 \times 10^2$                      |
|          | NOGWO     | $1.92 \times 10^3$                      | $8.62 \times 10^2$                      | $4.04 \times 10^3$                      | $7.38 \times 10^2$                      |
| F5       | GWO       | $7.56 \times 10^2$                      | $7.32 \times 10^1$                      | $9.97 \times 10^2$                      | $6.70 \times 10^2$                      |
|          | OGWO      | $7.54 \times 10^2$                      | $3.54 \times 10^1$                      | $8.30 \times 10^2$                      | $6.69 \times 10^2$                      |
|          | newOGWO   | $7.43 \times 10^2$                      | $3.36 \times 10^1$                      | $8.02 \times 10^2$                      | $6.53 \times 10^2$                      |
|          | NOGWO     | $8.57 \times 10^2$                      | $7.28 \times 10^1$                      | $1.05 \times 10^3$                      | $7.14 \times 10^2$                      |

Table 2. Cont.

| Function | Algorithm | Mean                                 | Std                                  | Max                                  | Min                                  |
|----------|-----------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| F6       | GWO       | $6.26 \times 10^2$                   | $3.82 \times 10^0$                   | $6.33 \times 10^2$                   | $6.17 \times 10^2$                   |
|          | OGWO      | $6.26 \times 10^2$                   | $6.62 \times 10^0$                   | $6.39 \times 10^2$                   | $6.12 \times 10^2$                   |
|          | newOGWO   | $6.25 \times 10^2$                   | $4.74 \times 10^0$                   | $6.34 \times 10^2$                   | $6.16 \times 10^2$                   |
|          | NOGWO     | $6.30 \times 10^2$                   | $7.76 \times 10^0$                   | $6.48 \times 10^2$                   | $6.19 \times 10^2$                   |
| F7       | GWO       | $1.16 \times 10^3$                   | $9.44 \times 10^1$                   | $1.42 \times 10^3$                   | $1.02 \times 10^3$                   |
|          | OGWO      | $1.15 \times 10^3$                   | $7.77 \times 10^1$                   | $1.31 \times 10^3$                   | $1.00 \times 10^3$                   |
|          | newOGWO   | $1.18 \times 10^3$                   | $9.21 \times 10^1$                   | $1.37 \times 10^3$                   | $1.02 \times 10^3$                   |
|          | NOGWO     | $1.21 \times 10^3$                   | $8.13 \times 10^1$                   | $1.38 \times 10^3$                   | $1.03 \times 10^3$                   |
| F8       | GWO       | $1.06 \times 10^3$                   | $3.51 \times 10^1$                   | $1.14 \times 10^3$                   | $9.96 \times 10^2$                   |
|          | OGWO      | $1.04 \times 10^3$                   | $4.13 \times 10^1$                   | $1.11 \times 10^3$                   | $9.72 \times 10^2$                   |
|          | newOGWO   | $1.06 \times 10^3$                   | $4.88 \times 10^1$                   | $1.23 \times 10^3$                   | $9.94 \times 10^2$                   |
|          | NOGWO     | $1.11 \times 10^3$                   | $7.56 \times 10^1$                   | $1.30 \times 10^3$                   | $9.76 \times 10^2$                   |
| F9       | GWO       | $1.17 \times 10^4$                   | $4.60 \times 10^3$                   | $2.45 \times 10^4$                   | $5.24 \times 10^3$                   |
|          | OGWO      | $1.08 \times 10^4$                   | $4.68 \times 10^3$                   | $2.37 \times 10^4$                   | $5.19 \times 10^3$                   |
|          | newOGWO   | $1.18 \times 10^4$                   | $4.49 \times 10^3$                   | $2.03 \times 10^4$                   | $5.79 \times 10^3$                   |
|          | NOGWO     | $1.69 \times 10^4$                   | $5.48 \times 10^3$                   | $2.92 \times 10^4$                   | $5.82 \times 10^3$                   |
| F10      | GWO       | $9.03 \times 10^3$                   | $2.43 \times 10^3$                   | $1.55 \times 10^4$                   | $6.59 \times 10^3$                   |
|          | OGWO      | $1.12 \times 10^4$                   | $2.98 \times 10^3$                   | $1.55 \times 10^4$                   | $6.98 \times 10^3$                   |
|          | newOGWO   | $1.13 \times 10^4$                   | $3.08 \times 10^3$                   | $1.55 \times 10^4$                   | $5.93 \times 10^3$                   |
|          | NOGWO     | $1.25 \times 10^4$                   | $3.37 \times 10^3$                   | $1.57 \times 10^4$                   | $6.64 \times 10^3$                   |
| F11      | GWO       | $7.28 \times 10^3$                   | $2.99 \times 10^3$                   | $1.47 \times 10^4$                   | $3.24 \times 10^3$                   |
|          | OGWO      | $6.11 \times 10^3$                   | $2.65 \times 10^3$                   | $1.25 \times 10^4$                   | $2.13 \times 10^3$                   |
|          | newOGWO   | $6.07 \times 10^3$                   | $1.95 \times 10^3$                   | $1.03 \times 10^4$                   | $1.87 \times 10^3$                   |
|          | NOGWO     | <b><math>2.35 \times 10^3</math></b> | <b><math>4.08 \times 10^2</math></b> | <b><math>3.36 \times 10^3</math></b> | <b><math>1.74 \times 10^3</math></b> |
| F12      | GWO       | $1.51 \times 10^9$                   | $1.65 \times 10^9$                   | $6.75 \times 10^9$                   | $7.67 \times 10^7$                   |
|          | OGWO      | $1.32 \times 10^9$                   | $1.40 \times 10^9$                   | $6.62 \times 10^9$                   | $6.80 \times 10^7$                   |
|          | newOGWO   | $1.42 \times 10^9$                   | $1.58 \times 10^9$                   | $6.39 \times 10^9$                   | $6.62 \times 10^7$                   |
|          | NOGWO     | $1.72 \times 10^9$                   | $1.60 \times 10^9$                   | $7.88 \times 10^9$                   | $1.90 \times 10^8$                   |
| F13      | GWO       | $2.17 \times 10^8$                   | $1.83 \times 10^8$                   | $8.25 \times 10^8$                   | $8.29 \times 10^6$                   |
|          | OGWO      | $2.04 \times 10^8$                   | $1.67 \times 10^8$                   | $7.66 \times 10^8$                   | $5.68 \times 10^6$                   |
|          | newOGWO   | $4.74 \times 10^8$                   | $1.35 \times 10^9$                   | $7.67 \times 10^9$                   | $3.42 \times 10^6$                   |
|          | NOGWO     | $2.56 \times 10^8$                   | $2.99 \times 10^8$                   | $1.28 \times 10^9$                   | <b><math>8.53 \times 10^5</math></b> |
| F14      | GWO       | $2.86 \times 10^6$                   | $3.89 \times 10^6$                   | $1.92 \times 10^7$                   | $7.65 \times 10^4$                   |
|          | OGWO      | $1.63 \times 10^6$                   | $1.61 \times 10^6$                   | $6.86 \times 10^6$                   | $1.23 \times 10^5$                   |
|          | newOGWO   | $1.67 \times 10^6$                   | $2.08 \times 10^6$                   | $1.00 \times 10^7$                   | $1.42 \times 10^5$                   |
|          | NOGWO     | <b><math>8.94 \times 10^5</math></b> | <b><math>9.41 \times 10^5</math></b> | <b><math>4.93 \times 10^6</math></b> | <b><math>9.51 \times 10^4</math></b> |
| F15      | GWO       | $6.64 \times 10^7$                   | $1.63 \times 10^8$                   | $7.13 \times 10^8$                   | $4.23 \times 10^4$                   |
|          | OGWO      | $2.29 \times 10^7$                   | $3.48 \times 10^7$                   | $1.15 \times 10^8$                   | $4.54 \times 10^4$                   |
|          | newOGWO   | $5.86 \times 10^7$                   | $1.27 \times 10^8$                   | $6.25 \times 10^8$                   | $5.22 \times 10^4$                   |
|          | NOGWO     | <b><math>1.54 \times 10^7</math></b> | <b><math>2.11 \times 10^7</math></b> | <b><math>7.06 \times 10^7</math></b> | <b><math>3.12 \times 10^4</math></b> |
| F16      | GWO       | $3.37 \times 10^3$                   | $4.00 \times 10^2$                   | $4.06 \times 10^3$                   | $2.74 \times 10^3$                   |
|          | OGWO      | $3.37 \times 10^3$                   | $4.50 \times 10^2$                   | $4.54 \times 10^3$                   | $2.58 \times 10^3$                   |
|          | newOGWO   | $3.42 \times 10^3$                   | $4.50 \times 10^2$                   | $4.78 \times 10^3$                   | $2.76 \times 10^3$                   |
|          | NOGWO     | $3.73 \times 10^3$                   | $7.50 \times 10^2$                   | $5.60 \times 10^3$                   | <b><math>2.47 \times 10^3</math></b> |
| F17      | GWO       | $3.09 \times 10^3$                   | $2.40 \times 10^2$                   | $3.89 \times 10^3$                   | $2.79 \times 10^3$                   |
|          | OGWO      | $3.06 \times 10^3$                   | $3.01 \times 10^2$                   | $3.79 \times 10^3$                   | $2.48 \times 10^3$                   |
|          | newOGWO   | $3.30 \times 10^3$                   | $5.08 \times 10^2$                   | $4.57 \times 10^3$                   | $2.55 \times 10^3$                   |
|          | NOGWO     | $3.23 \times 10^3$                   | $4.96 \times 10^2$                   | $4.38 \times 10^3$                   | $2.53 \times 10^3$                   |
| F18      | GWO       | $1.54 \times 10^7$                   | $2.00 \times 10^7$                   | $6.96 \times 10^7$                   | $7.51 \times 10^5$                   |
|          | OGWO      | $9.56 \times 10^6$                   | $1.09 \times 10^7$                   | $4.93 \times 10^7$                   | $1.31 \times 10^6$                   |
|          | newOGWO   | $1.29 \times 10^7$                   | $1.60 \times 10^7$                   | $6.40 \times 10^7$                   | $7.65 \times 10^5$                   |
|          | NOGWO     | <b><math>6.13 \times 10^6</math></b> | <b><math>9.00 \times 10^6</math></b> | <b><math>4.27 \times 10^7</math></b> | $1.04 \times 10^6$                   |

Table 2. Cont.

| Function              | Algorithm | Mean                                 | Std                                  | Max                                  | Min                                  |
|-----------------------|-----------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| F19                   | GWO       | $1.26 \times 10^7$                   | $2.99 \times 10^7$                   | $1.33 \times 10^8$                   | $2.04 \times 10^5$                   |
|                       | OGWO      | $9.25 \times 10^6$                   | $2.30 \times 10^7$                   | $1.24 \times 10^8$                   | $1.32 \times 10^5$                   |
|                       | newOGWO   | $9.46 \times 10^6$                   | $2.00 \times 10^7$                   | $8.91 \times 10^7$                   | $8.11 \times 10^4$                   |
|                       | NOGWO     | <b><math>7.47 \times 10^6</math></b> | <b><math>1.36 \times 10^7</math></b> | <b><math>6.48 \times 10^7</math></b> | $2.34 \times 10^5$                   |
| F20                   | GWO       | $3.11 \times 10^3$                   | $3.50 \times 10^2$                   | $4.17 \times 10^3$                   | $2.47 \times 10^3$                   |
|                       | OGWO      | $3.48 \times 10^3$                   | $5.48 \times 10^2$                   | $4.37 \times 10^3$                   | $2.55 \times 10^3$                   |
|                       | newOGWO   | $3.33 \times 10^3$                   | $5.06 \times 10^2$                   | $4.25 \times 10^3$                   | $2.48 \times 10^3$                   |
|                       | NOGWO     | $3.45 \times 10^3$                   | $5.87 \times 10^2$                   | $4.27 \times 10^3$                   | $2.56 \times 10^3$                   |
| F21                   | GWO       | $2.57 \times 10^3$                   | $7.73 \times 10^1$                   | $2.86 \times 10^3$                   | $2.46 \times 10^3$                   |
|                       | OGWO      | $2.56 \times 10^3$                   | $4.53 \times 10^1$                   | $2.74 \times 10^3$                   | $2.50 \times 10^3$                   |
|                       | newOGWO   | $2.56 \times 10^3$                   | $5.71 \times 10^1$                   | $2.80 \times 10^3$                   | $2.49 \times 10^3$                   |
|                       | NOGWO     | $2.59 \times 10^3$                   | $8.65 \times 10^1$                   | $2.85 \times 10^3$                   | $2.49 \times 10^3$                   |
| F22                   | GWO       | $1.05 \times 10^4$                   | $2.63 \times 10^3$                   | $1.76 \times 10^4$                   | $8.33 \times 10^3$                   |
|                       | OGWO      | $1.21 \times 10^4$                   | $2.93 \times 10^3$                   | $1.72 \times 10^4$                   | $4.78 \times 10^3$                   |
|                       | newOGWO   | $1.27 \times 10^4$                   | $3.13 \times 10^3$                   | $1.73 \times 10^4$                   | $8.75 \times 10^3$                   |
|                       | NOGWO     | <b><math>1.03 \times 10^4</math></b> | $4.68 \times 10^3$                   | <b><math>1.68 \times 10^4</math></b> | <b><math>2.98 \times 10^3</math></b> |
| F23                   | GWO       | $3.04 \times 10^3$                   | $7.87 \times 10^1$                   | $3.30 \times 10^3$                   | $2.91 \times 10^3$                   |
|                       | OGWO      | $3.06 \times 10^3$                   | $7.86 \times 10^1$                   | $3.28 \times 10^3$                   | $2.94 \times 10^3$                   |
|                       | newOGWO   | $3.06 \times 10^3$                   | $9.78 \times 10^1$                   | $3.34 \times 10^3$                   | $2.94 \times 10^3$                   |
|                       | NOGWO     | <b><math>3.03 \times 10^3</math></b> | <b><math>6.82 \times 10^1</math></b> | $3.29 \times 10^3$                   | <b><math>2.91 \times 10^3</math></b> |
| F24                   | GWO       | $3.24 \times 10^3$                   | $1.03 \times 10^2$                   | $3.50 \times 10^3$                   | $3.10 \times 10^3$                   |
|                       | OGWO      | $3.25 \times 10^3$                   | $9.89 \times 10^1$                   | $3.49 \times 10^3$                   | $3.12 \times 10^3$                   |
|                       | newOGWO   | $3.22 \times 10^3$                   | $1.09 \times 10^2$                   | $3.52 \times 10^3$                   | $3.11 \times 10^3$                   |
|                       | NOGWO     | <b><math>3.22 \times 10^3</math></b> | $1.21 \times 10^2$                   | <b><math>3.48 \times 10^3</math></b> | <b><math>3.05 \times 10^3</math></b> |
| F25                   | GWO       | $4.11 \times 10^3$                   | $5.96 \times 10^2$                   | $5.28 \times 10^3$                   | $3.33 \times 10^3$                   |
|                       | OGWO      | $3.86 \times 10^3$                   | $3.67 \times 10^2$                   | $4.95 \times 10^3$                   | $3.39 \times 10^3$                   |
|                       | newOGWO   | $3.92 \times 10^3$                   | $4.39 \times 10^2$                   | $5.60 \times 10^3$                   | $3.40 \times 10^3$                   |
|                       | NOGWO     | $4.09 \times 10^3$                   | $4.27 \times 10^2$                   | $5.31 \times 10^3$                   | $3.49 \times 10^3$                   |
| F26                   | GWO       | $7.18 \times 10^3$                   | $7.18 \times 10^2$                   | $9.27 \times 10^3$                   | $6.03 \times 10^3$                   |
|                       | OGWO      | $7.27 \times 10^3$                   | $6.09 \times 10^2$                   | $8.67 \times 10^3$                   | $6.02 \times 10^3$                   |
|                       | newOGWO   | $7.19 \times 10^3$                   | $6.95 \times 10^2$                   | $8.93 \times 10^3$                   | $6.17 \times 10^3$                   |
|                       | NOGWO     | <b><math>6.89 \times 10^3</math></b> | $8.29 \times 10^2$                   | $9.05 \times 10^3$                   | <b><math>5.47 \times 10^3</math></b> |
| F27                   | GWO       | $3.70 \times 10^3$                   | $1.09 \times 10^2$                   | $3.96 \times 10^3$                   | $3.47 \times 10^3$                   |
|                       | OGWO      | $3.73 \times 10^3$                   | $1.16 \times 10^2$                   | $4.01 \times 10^3$                   | $3.52 \times 10^3$                   |
|                       | newOGWO   | $3.71 \times 10^3$                   | $1.20 \times 10^2$                   | $4.01 \times 10^3$                   | $3.50 \times 10^3$                   |
|                       | NOGWO     | <b><math>3.69 \times 10^3</math></b> | <b><math>9.97 \times 10^1</math></b> | <b><math>3.95 \times 10^3</math></b> | $3.52 \times 10^3$                   |
| F28                   | GWO       | $4.68 \times 10^3$                   | $6.92 \times 10^2$                   | $6.74 \times 10^3$                   | $3.69 \times 10^3$                   |
|                       | OGWO      | $4.72 \times 10^3$                   | $4.63 \times 10^2$                   | $5.52 \times 10^3$                   | $3.79 \times 10^3$                   |
|                       | newOGWO   | $4.74 \times 10^3$                   | $4.68 \times 10^2$                   | $5.87 \times 10^3$                   | $3.71 \times 10^3$                   |
|                       | NOGWO     | <b><math>4.65 \times 10^3</math></b> | <b><math>3.53 \times 10^2</math></b> | <b><math>5.36 \times 10^3</math></b> | $4.10 \times 10^3$                   |
| F29                   | GWO       | $5.08 \times 10^3$                   | $4.98 \times 10^2$                   | $6.29 \times 10^3$                   | $4.34 \times 10^3$                   |
|                       | OGWO      | $4.94 \times 10^3$                   | $3.37 \times 10^2$                   | $5.59 \times 10^3$                   | $4.14 \times 10^3$                   |
|                       | newOGWO   | $5.11 \times 10^3$                   | $4.21 \times 10^2$                   | $5.99 \times 10^3$                   | $4.43 \times 10^3$                   |
|                       | NOGWO     | <b><math>4.86 \times 10^3</math></b> | $3.55 \times 10^2$                   | $5.75 \times 10^3$                   | $4.22 \times 10^3$                   |
| F30                   | GWO       | $1.84 \times 10^8$                   | $1.11 \times 10^8$                   | $6.56 \times 10^8$                   | $5.95 \times 10^7$                   |
|                       | OGWO      | $1.62 \times 10^8$                   | $6.01 \times 10^7$                   | $3.11 \times 10^8$                   | $7.06 \times 10^7$                   |
|                       | newOGWO   | $1.62 \times 10^8$                   | $3.82 \times 10^7$                   | $2.48 \times 10^8$                   | $8.84 \times 10^7$                   |
|                       | NOGWO     | $2.01 \times 10^8$                   | $5.95 \times 10^7$                   | $3.41 \times 10^8$                   | $1.17 \times 10^8$                   |
| Data analysis results |           | 14/30                                | 9/30                                 | 11/30                                | 10/30                                |

#### 4.3. Comparison with Other Outstanding Swarm Intelligence Algorithms

To comprehensively verify the NOGWO's excellent competence, four excellent algorithms, WOA [18], DE [19], ALO [20] and SCA [21], are selected for comparison. Six swarm intelligence algorithms are tested on 30 CEC2017 test functions. Table 3 presents the experimental outcomes. Moreover, Table 4 displays the win (W), tie (T) and lose (L) of each swarm intelligence algorithm on four evaluation indicators. Figure 3 corresponds to the data presented in Table 4, wherein the values in the bar represent the count of wins for each algorithm across different evaluation indicators.

**Table 3.** The Accuracy test of NOGWO.

| Function | Algorithm | Mean                  | Std                   | Max                   | Min                   |
|----------|-----------|-----------------------|-----------------------|-----------------------|-----------------------|
| F1       | NOGWO     | $3.17 \times 10^9$    | $2.12 \times 10^9$    | $8.56 \times 10^9$    | $1.25 \times 10^8$    |
|          | GWO       | $2.91 \times 10^9$    | $2.02 \times 10^9$    | $7.91 \times 10^9$    | $4.40 \times 10^8$    |
|          | DE        | $1.08 \times 10^6$    | $6.06 \times 10^5$    | $2.67 \times 10^6$    | $5.09 \times 10^4$    |
|          | WOA       | $1.79 \times 10^9$    | $1.31 \times 10^9$    | $4.90 \times 10^9$    | $3.53 \times 10^7$    |
|          | ALO       | $3.72 \times 10^4$    | $7.50 \times 10^4$    | $4.28 \times 10^5$    | $1.33 \times 10^3$    |
|          | SCA       | $2.18 \times 10^{10}$ | $4.06 \times 10^9$    | $3.18 \times 10^{10}$ | $1.26 \times 10^{10}$ |
| F2       | NOGWO     | $2.78 \times 10^{32}$ | $1.04 \times 10^{33}$ | $5.46 \times 10^{33}$ | $2.14 \times 10^{21}$ |
|          | GWO       | $1.94 \times 10^{34}$ | $1.05 \times 10^{35}$ | $5.85 \times 10^{35}$ | $3.24 \times 10^{19}$ |
|          | DE        | $1.41 \times 10^{32}$ | $2.65 \times 10^{32}$ | $1.03 \times 10^{33}$ | $7.91 \times 10^{27}$ |
|          | WOA       | $5.18 \times 10^{30}$ | $2.77 \times 10^{31}$ | $1.55 \times 10^{32}$ | $2.10 \times 10^{17}$ |
|          | ALO       | $8.34 \times 10^{21}$ | $4.51 \times 10^{22}$ | $2.51 \times 10^{23}$ | $1.90 \times 10^{13}$ |
|          | SCA       | $2.32 \times 10^{38}$ | $9.02 \times 10^{38}$ | $4.93 \times 10^{39}$ | $3.25 \times 10^{33}$ |
| F3       | NOGWO     | $4.45 \times 10^4$    | $7.54 \times 10^3$    | $5.82 \times 10^4$    | $2.78 \times 10^4$    |
|          | GWO       | $6.35 \times 10^4$    | $1.20 \times 10^4$    | $8.64 \times 10^4$    | $3.48 \times 10^4$    |
|          | DE        | $1.62 \times 10^5$    | $2.76 \times 10^4$    | $2.23 \times 10^5$    | $1.06 \times 10^5$    |
|          | WOA       | $1.01 \times 10^5$    | $3.18 \times 10^4$    | $1.70 \times 10^5$    | $4.94 \times 10^4$    |
|          | ALO       | $2.21 \times 10^5$    | $4.69 \times 10^4$    | $2.96 \times 10^5$    | $1.27 \times 10^5$    |
|          | SCA       | $8.51 \times 10^4$    | $1.66 \times 10^4$    | $1.41 \times 10^5$    | $5.71 \times 10^4$    |
| F4       | NOGWO     | $6.25 \times 10^2$    | $5.40 \times 10^1$    | $7.36 \times 10^2$    | $5.20 \times 10^2$    |
|          | GWO       | $6.70 \times 10^2$    | $1.17 \times 10^2$    | $1.13 \times 10^3$    | $5.56 \times 10^2$    |
|          | DE        | $5.20 \times 10^2$    | $1.92 \times 10^1$    | $5.54 \times 10^2$    | $4.98 \times 10^2$    |
|          | WOA       | $6.28 \times 10^2$    | $9.41 \times 10^1$    | $9.79 \times 10^2$    | $5.18 \times 10^2$    |
|          | ALO       | $5.42 \times 10^2$    | $3.17 \times 10^1$    | $6.12 \times 10^2$    | $4.79 \times 10^2$    |
|          | SCA       | $3.10 \times 10^3$    | $8.73 \times 10^2$    | $5.28 \times 10^3$    | $1.92 \times 10^3$    |
| F5       | NOGWO     | $6.69 \times 10^2$    | $4.55 \times 10^1$    | $7.96 \times 10^2$    | $5.94 \times 10^2$    |
|          | GWO       | $6.11 \times 10^2$    | $2.15 \times 10^1$    | $6.43 \times 10^2$    | $5.72 \times 10^2$    |
|          | DE        | $6.99 \times 10^2$    | $1.29 \times 10^1$    | $7.30 \times 10^2$    | $6.64 \times 10^2$    |
|          | WOA       | $7.32 \times 10^2$    | $5.03 \times 10^1$    | $8.52 \times 10^2$    | $6.41 \times 10^2$    |
|          | ALO       | $6.76 \times 10^2$    | $5.32 \times 10^1$    | $7.86 \times 10^2$    | $5.89 \times 10^2$    |
|          | SCA       | $8.28 \times 10^2$    | $2.82 \times 10^1$    | $8.84 \times 10^2$    | $7.78 \times 10^2$    |
| F6       | NOGWO     | $6.14 \times 10^2$    | $6.10 \times 10^0$    | $6.31 \times 10^2$    | $6.05 \times 10^2$    |
|          | GWO       | $6.12 \times 10^2$    | $4.07 \times 10^0$    | $6.23 \times 10^2$    | $6.05 \times 10^2$    |
|          | DE        | $6.00 \times 10^2$    | $4.08 \times 10^{-2}$ | $6.00 \times 10^2$    | $6.00 \times 10^2$    |
|          | WOA       | $6.50 \times 10^2$    | $1.13 \times 10^1$    | $6.72 \times 10^2$    | $6.32 \times 10^2$    |
|          | ALO       | $6.49 \times 10^2$    | $1.03 \times 10^1$    | $6.67 \times 10^2$    | $6.23 \times 10^2$    |
|          | SCA       | $6.63 \times 10^2$    | $7.15 \times 10^0$    | $6.82 \times 10^2$    | $6.50 \times 10^2$    |
| F7       | NOGWO     | $9.21 \times 10^2$    | $5.80 \times 10^1$    | $1.06 \times 10^3$    | $8.45 \times 10^2$    |
|          | GWO       | $9.08 \times 10^2$    | $5.61 \times 10^1$    | $1.10 \times 10^3$    | $8.33 \times 10^2$    |
|          | DE        | $9.30 \times 10^2$    | $1.20 \times 10^1$    | $9.48 \times 10^2$    | $9.02 \times 10^2$    |
|          | WOA       | $1.02 \times 10^3$    | $6.24 \times 10^1$    | $1.17 \times 10^3$    | $9.39 \times 10^2$    |
|          | ALO       | $1.14 \times 10^3$    | $8.60 \times 10^1$    | $1.32 \times 10^3$    | $9.66 \times 10^2$    |
|          | SCA       | $1.26 \times 10^3$    | $7.05 \times 10^1$    | $1.44 \times 10^3$    | $1.16 \times 10^3$    |

Table 3. Cont.

| Function | Algorithm | Mean                                 | Std                                  | Max                                  | Min                                  |
|----------|-----------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| F8       | NOGWO     | $9.16 \times 10^2$                   | $3.30 \times 10^1$                   | $1.06 \times 10^3$                   | $8.68 \times 10^2$                   |
|          | GWO       | $9.07 \times 10^2$                   | $3.94 \times 10^1$                   | $1.04 \times 10^3$                   | $8.53 \times 10^2$                   |
|          | DE        | $9.94 \times 10^2$                   | $1.28 \times 10^1$                   | $1.02 \times 10^3$                   | $9.59 \times 10^2$                   |
|          | WOA       | $1.00 \times 10^3$                   | $4.38 \times 10^1$                   | $1.10 \times 10^3$                   | $9.22 \times 10^2$                   |
|          | ALO       | $9.46 \times 10^2$                   | $3.90 \times 10^1$                   | $1.09 \times 10^3$                   | $8.93 \times 10^2$                   |
|          | SCA       | $1.10 \times 10^3$                   | $2.65 \times 10^1$                   | $1.15 \times 10^3$                   | $1.03 \times 10^3$                   |
| F9       | NOGWO     | $2.96 \times 10^3$                   | $1.25 \times 10^3$                   | $6.56 \times 10^3$                   | <b><math>1.22 \times 10^3</math></b> |
|          | GWO       | $2.91 \times 10^3$                   | $1.08 \times 10^3$                   | $5.88 \times 10^3$                   | $1.37 \times 10^3$                   |
|          | DE        | $2.01 \times 10^3$                   | $4.17 \times 10^2$                   | $2.91 \times 10^3$                   | $1.30 \times 10^3$                   |
|          | WOA       | $6.39 \times 10^3$                   | $2.51 \times 10^3$                   | $1.24 \times 10^4$                   | $2.82 \times 10^3$                   |
|          | ALO       | $5.05 \times 10^3$                   | $1.36 \times 10^3$                   | $8.56 \times 10^3$                   | $2.80 \times 10^3$                   |
|          | SCA       | $8.53 \times 10^3$                   | $1.85 \times 10^3$                   | $1.15 \times 10^4$                   | $4.98 \times 10^3$                   |
| F10      | NOGWO     | $7.78 \times 10^3$                   | $1.73 \times 10^3$                   | $9.21 \times 10^3$                   | $4.06 \times 10^3$                   |
|          | GWO       | $5.46 \times 10^3$                   | $1.77 \times 10^3$                   | $9.44 \times 10^3$                   | $3.30 \times 10^3$                   |
|          | DE        | $7.67 \times 10^3$                   | $3.42 \times 10^2$                   | $8.24 \times 10^3$                   | $6.61 \times 10^3$                   |
|          | WOA       | $5.98 \times 10^3$                   | $6.89 \times 10^2$                   | $7.11 \times 10^3$                   | $4.11 \times 10^3$                   |
|          | ALO       | $5.62 \times 10^3$                   | $8.67 \times 10^2$                   | $7.33 \times 10^3$                   | $3.81 \times 10^3$                   |
|          | SCA       | $8.78 \times 10^3$                   | $3.45 \times 10^2$                   | $9.34 \times 10^3$                   | $8.10 \times 10^3$                   |
| F11      | NOGWO     | <b><math>1.43 \times 10^3</math></b> | <b><math>1.19 \times 10^2</math></b> | <b><math>1.75 \times 10^3</math></b> | <b><math>1.27 \times 10^3</math></b> |
|          | GWO       | $2.72 \times 10^3$                   | $1.32 \times 10^3$                   | $7.03 \times 10^3$                   | $1.35 \times 10^3$                   |
|          | DE        | $1.95 \times 10^3$                   | $4.88 \times 10^2$                   | $3.55 \times 10^3$                   | $1.38 \times 10^3$                   |
|          | WOA       | $1.88 \times 10^3$                   | $8.22 \times 10^2$                   | $4.44 \times 10^3$                   | $1.35 \times 10^3$                   |
|          | ALO       | $1.58 \times 10^3$                   | $2.22 \times 10^2$                   | $2.30 \times 10^3$                   | $1.34 \times 10^3$                   |
|          | SCA       | $4.00 \times 10^3$                   | $1.03 \times 10^3$                   | $6.84 \times 10^3$                   | $2.53 \times 10^3$                   |
| F12      | NOGWO     | $1.04 \times 10^8$                   | $9.24 \times 10^7$                   | $4.36 \times 10^8$                   | $4.49 \times 10^6$                   |
|          | GWO       | $2.16 \times 10^8$                   | $5.53 \times 10^8$                   | $3.12 \times 10^9$                   | $1.04 \times 10^7$                   |
|          | DE        | $3.09 \times 10^7$                   | $1.54 \times 10^7$                   | $8.32 \times 10^7$                   | $1.06 \times 10^7$                   |
|          | WOA       | $6.16 \times 10^7$                   | $5.16 \times 10^7$                   | $2.20 \times 10^8$                   | $1.91 \times 10^6$                   |
|          | ALO       | $2.94 \times 10^7$                   | $2.47 \times 10^7$                   | $9.26 \times 10^7$                   | $2.24 \times 10^6$                   |
|          | SCA       | $2.62 \times 10^9$                   | $6.81 \times 10^8$                   | $4.32 \times 10^9$                   | $1.31 \times 10^9$                   |
| F13      | NOGWO     | $1.10 \times 10^7$                   | $3.86 \times 10^7$                   | $2.06 \times 10^8$                   | <b><math>2.00 \times 10^4</math></b> |
|          | GWO       | $2.09 \times 10^7$                   | $5.83 \times 10^7$                   | $2.97 \times 10^8$                   | $8.74 \times 10^4$                   |
|          | DE        | $4.32 \times 10^6$                   | $4.34 \times 10^6$                   | $2.16 \times 10^7$                   | $4.48 \times 10^5$                   |
|          | WOA       | $2.19 \times 10^5$                   | $1.58 \times 10^5$                   | $7.17 \times 10^5$                   | $3.23 \times 10^4$                   |
|          | ALO       | $1.24 \times 10^5$                   | $8.28 \times 10^4$                   | $4.63 \times 10^5$                   | $2.35 \times 10^4$                   |
|          | SCA       | $1.23 \times 10^9$                   | $8.59 \times 10^8$                   | $4.96 \times 10^9$                   | $3.56 \times 10^8$                   |
| F14      | NOGWO     | <b><math>4.11 \times 10^5</math></b> | $6.05 \times 10^5$                   | $2.58 \times 10^6$                   | $9.83 \times 10^3$                   |
|          | GWO       | $8.97 \times 10^5$                   | $1.14 \times 10^6$                   | $4.26 \times 10^6$                   | $7.37 \times 10^3$                   |
|          | DE        | $4.52 \times 10^5$                   | $2.49 \times 10^5$                   | $1.20 \times 10^6$                   | $1.54 \times 10^5$                   |
|          | WOA       | $5.47 \times 10^5$                   | $6.13 \times 10^5$                   | $2.62 \times 10^6$                   | $3.68 \times 10^3$                   |
|          | ALO       | $4.88 \times 10^5$                   | $4.81 \times 10^5$                   | $1.86 \times 10^6$                   | $1.91 \times 10^4$                   |
|          | SCA       | $9.54 \times 10^5$                   | $7.96 \times 10^5$                   | $3.95 \times 10^6$                   | $1.85 \times 10^5$                   |
| F15      | NOGWO     | $6.95 \times 10^5$                   | $1.43 \times 10^6$                   | $5.88 \times 10^6$                   | $1.48 \times 10^4$                   |
|          | GWO       | $1.09 \times 10^6$                   | $1.71 \times 10^6$                   | $6.17 \times 10^6$                   | $5.14 \times 10^3$                   |
|          | DE        | $9.32 \times 10^5$                   | $6.59 \times 10^5$                   | $3.21 \times 10^6$                   | $1.68 \times 10^5$                   |
|          | WOA       | $1.00 \times 10^5$                   | $1.51 \times 10^5$                   | $8.29 \times 10^5$                   | $1.67 \times 10^4$                   |
|          | ALO       | $4.62 \times 10^4$                   | $3.03 \times 10^4$                   | $1.48 \times 10^5$                   | $5.80 \times 10^3$                   |
|          | SCA       | $5.46 \times 10^7$                   | $5.94 \times 10^7$                   | $2.08 \times 10^8$                   | $2.46 \times 10^6$                   |
| F16      | NOGWO     | $2.92 \times 10^3$                   | $4.76 \times 10^2$                   | $3.96 \times 10^3$                   | $2.23 \times 10^3$                   |
|          | GWO       | $2.64 \times 10^3$                   | $3.37 \times 10^2$                   | $3.34 \times 10^3$                   | $2.10 \times 10^3$                   |
|          | DE        | $3.02 \times 10^3$                   | $1.73 \times 10^2$                   | $3.45 \times 10^3$                   | $2.64 \times 10^3$                   |
|          | WOA       | $3.08 \times 10^3$                   | $4.47 \times 10^2$                   | $4.39 \times 10^3$                   | $2.27 \times 10^3$                   |
|          | ALO       | $3.27 \times 10^3$                   | $3.11 \times 10^2$                   | $3.98 \times 10^3$                   | $2.48 \times 10^3$                   |
|          | SCA       | $4.10 \times 10^3$                   | $2.67 \times 10^2$                   | $4.68 \times 10^3$                   | $3.40 \times 10^3$                   |

Table 3. Cont.

| Function | Algorithm | Mean               | Std                | Max                | Min                |
|----------|-----------|--------------------|--------------------|--------------------|--------------------|
| F17      | NOGWO     | $2.07 \times 10^3$ | $1.90 \times 10^2$ | $2.64 \times 10^3$ | $1.80 \times 10^3$ |
|          | GWO       | $2.10 \times 10^3$ | $2.08 \times 10^2$ | $2.51 \times 10^3$ | $1.84 \times 10^3$ |
|          | DE        | $2.25 \times 10^3$ | $1.07 \times 10^2$ | $2.44 \times 10^3$ | $2.00 \times 10^3$ |
|          | WOA       | $2.47 \times 10^3$ | $2.55 \times 10^2$ | $2.85 \times 10^3$ | $1.93 \times 10^3$ |
|          | ALO       | $2.57 \times 10^3$ | $2.30 \times 10^2$ | $3.15 \times 10^3$ | $2.10 \times 10^3$ |
|          | SCA       | $2.77 \times 10^3$ | $2.13 \times 10^2$ | $3.23 \times 10^3$ | $2.33 \times 10^3$ |
| F18      | NOGWO     | $1.03 \times 10^6$ | $1.04 \times 10^6$ | $4.39 \times 10^6$ | $7.88 \times 10^4$ |
|          | GWO       | $1.20 \times 10^6$ | $1.42 \times 10^6$ | $5.19 \times 10^6$ | $8.46 \times 10^4$ |
|          | DE        | $3.54 \times 10^6$ | $1.74 \times 10^6$ | $6.94 \times 10^6$ | $5.47 \times 10^5$ |
|          | WOA       | $3.30 \times 10^6$ | $3.48 \times 10^6$ | $1.48 \times 10^7$ | $1.40 \times 10^5$ |
|          | ALO       | $1.42 \times 10^6$ | $1.38 \times 10^6$ | $5.19 \times 10^6$ | $1.31 \times 10^5$ |
|          | SCA       | $1.30 \times 10^7$ | $8.97 \times 10^6$ | $4.57 \times 10^7$ | $2.82 \times 10^6$ |
| F19      | NOGWO     | $1.27 \times 10^6$ | $2.19 \times 10^6$ | $1.11 \times 10^7$ | $9.90 \times 10^3$ |
|          | GWO       | $9.89 \times 10^5$ | $1.18 \times 10^6$ | $5.14 \times 10^6$ | $9.02 \times 10^3$ |
|          | DE        | $5.89 \times 10^5$ | $5.26 \times 10^5$ | $2.51 \times 10^6$ | $1.71 \times 10^5$ |
|          | WOA       | $4.47 \times 10^5$ | $9.22 \times 10^5$ | $4.33 \times 10^6$ | $6.52 \times 10^3$ |
|          | ALO       | $4.22 \times 10^6$ | $2.67 \times 10^6$ | $1.02 \times 10^7$ | $2.14 \times 10^4$ |
|          | SCA       | $1.16 \times 10^8$ | $6.48 \times 10^7$ | $2.98 \times 10^8$ | $3.51 \times 10^7$ |
| F20      | NOGWO     | $2.43 \times 10^3$ | $2.10 \times 10^2$ | $2.92 \times 10^3$ | $2.23 \times 10^3$ |
|          | GWO       | $2.50 \times 10^3$ | $2.09 \times 10^2$ | $3.06 \times 10^3$ | $2.16 \times 10^3$ |
|          | DE        | $2.52 \times 10^3$ | $1.04 \times 10^2$ | $2.76 \times 10^3$ | $2.26 \times 10^3$ |
|          | WOA       | $2.71 \times 10^3$ | $2.15 \times 10^2$ | $3.07 \times 10^3$ | $2.34 \times 10^3$ |
|          | ALO       | $2.74 \times 10^3$ | $2.20 \times 10^2$ | $3.24 \times 10^3$ | $2.36 \times 10^3$ |
|          | SCA       | $2.96 \times 10^3$ | $1.29 \times 10^2$ | $3.21 \times 10^3$ | $2.69 \times 10^3$ |
| F21      | NOGWO     | $2.40 \times 10^3$ | $4.52 \times 10^1$ | $2.52 \times 10^3$ | $2.25 \times 10^3$ |
|          | GWO       | $2.41 \times 10^3$ | $2.91 \times 10^1$ | $2.48 \times 10^3$ | $2.37 \times 10^3$ |
|          | DE        | $2.50 \times 10^3$ | $1.47 \times 10^1$ | $2.52 \times 10^3$ | $2.46 \times 10^3$ |
|          | WOA       | $2.50 \times 10^3$ | $3.92 \times 10^1$ | $2.58 \times 10^3$ | $2.42 \times 10^3$ |
|          | ALO       | $2.44 \times 10^3$ | $2.96 \times 10^1$ | $2.53 \times 10^3$ | $2.39 \times 10^3$ |
|          | SCA       | $2.60 \times 10^3$ | $2.10 \times 10^1$ | $2.65 \times 10^3$ | $2.54 \times 10^3$ |
| F22      | NOGWO     | $2.79 \times 10^3$ | $2.89 \times 10^2$ | $3.85 \times 10^3$ | $2.46 \times 10^3$ |
|          | GWO       | $5.63 \times 10^3$ | $2.18 \times 10^3$ | $1.06 \times 10^4$ | $2.65 \times 10^3$ |
|          | DE        | $5.98 \times 10^3$ | $1.98 \times 10^3$ | $9.83 \times 10^3$ | $3.70 \times 10^3$ |
|          | WOA       | $6.91 \times 10^3$ | $1.62 \times 10^3$ | $8.65 \times 10^3$ | $2.56 \times 10^3$ |
|          | ALO       | $5.27 \times 10^3$ | $2.43 \times 10^3$ | $9.04 \times 10^3$ | $2.31 \times 10^3$ |
|          | SCA       | $1.00 \times 10^4$ | $1.20 \times 10^3$ | $1.07 \times 10^4$ | $4.86 \times 10^3$ |
| F23      | NOGWO     | $2.81 \times 10^3$ | $6.66 \times 10^1$ | $2.92 \times 10^3$ | $2.72 \times 10^3$ |
|          | GWO       | $2.81 \times 10^3$ | $6.46 \times 10^1$ | $3.01 \times 10^3$ | $2.72 \times 10^3$ |
|          | DE        | $2.84 \times 10^3$ | $1.68 \times 10^1$ | $2.86 \times 10^3$ | $2.79 \times 10^3$ |
|          | WOA       | $2.89 \times 10^3$ | $4.61 \times 10^1$ | $2.99 \times 10^3$ | $2.81 \times 10^3$ |
|          | ALO       | $2.86 \times 10^3$ | $5.62 \times 10^1$ | $2.98 \times 10^3$ | $2.73 \times 10^3$ |
|          | SCA       | $3.10 \times 10^3$ | $5.12 \times 10^1$ | $3.22 \times 10^3$ | $2.98 \times 10^3$ |
| F24      | NOGWO     | $2.96 \times 10^3$ | $6.18 \times 10^1$ | $3.08 \times 10^3$ | $2.88 \times 10^3$ |
|          | GWO       | $2.94 \times 10^3$ | $3.64 \times 10^1$ | $3.05 \times 10^3$ | $2.89 \times 10^3$ |
|          | DE        | $3.04 \times 10^3$ | $1.45 \times 10^1$ | $3.06 \times 10^3$ | $3.01 \times 10^3$ |
|          | WOA       | $3.02 \times 10^3$ | $6.31 \times 10^1$ | $3.24 \times 10^3$ | $2.94 \times 10^3$ |
|          | ALO       | $3.01 \times 10^3$ | $4.27 \times 10^1$ | $3.08 \times 10^3$ | $2.93 \times 10^3$ |
|          | SCA       | $3.26 \times 10^3$ | $4.30 \times 10^1$ | $3.34 \times 10^3$ | $3.16 \times 10^3$ |
| F25      | NOGWO     | $3.03 \times 10^3$ | $6.45 \times 10^1$ | $3.24 \times 10^3$ | $2.93 \times 10^3$ |
|          | GWO       | $3.04 \times 10^3$ | $8.52 \times 10^1$ | $3.29 \times 10^3$ | $2.95 \times 10^3$ |
|          | DE        | $2.90 \times 10^3$ | $9.85 \times 10^0$ | $2.94 \times 10^3$ | $2.89 \times 10^3$ |
|          | WOA       | $3.02 \times 10^3$ | $4.23 \times 10^1$ | $3.13 \times 10^3$ | $2.93 \times 10^3$ |
|          | ALO       | $2.97 \times 10^3$ | $3.34 \times 10^1$ | $3.07 \times 10^3$ | $2.92 \times 10^3$ |
|          | SCA       | $3.65 \times 10^3$ | $2.82 \times 10^2$ | $4.58 \times 10^3$ | $3.34 \times 10^3$ |



Table 3. Cont.

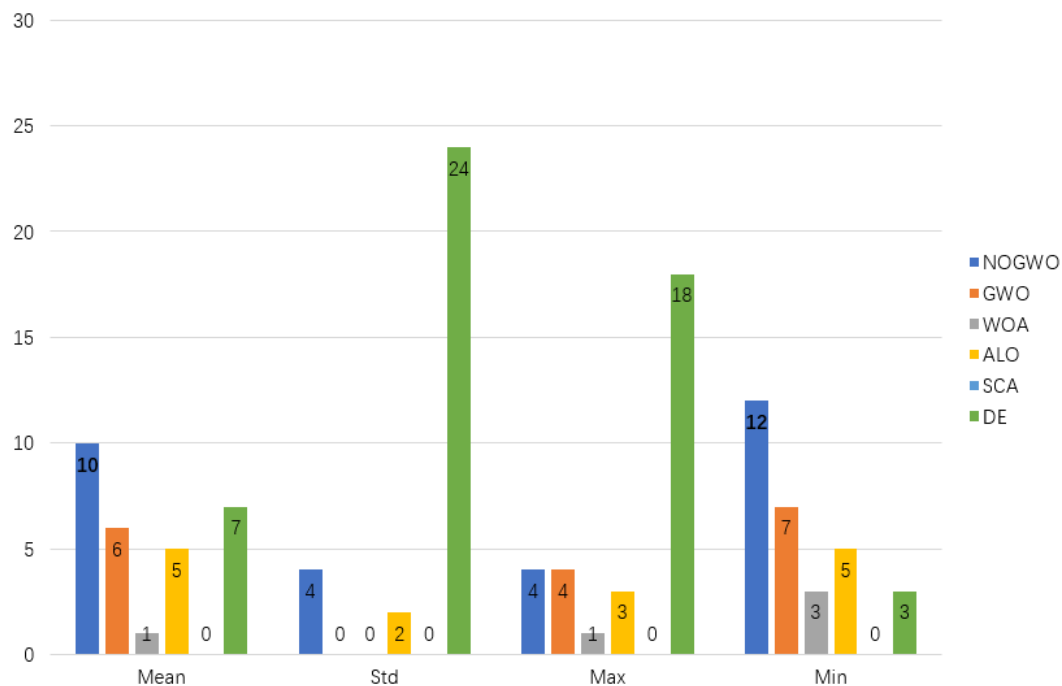
| Function | Algorithm | Mean               | Std                | Max                | Min                |
|----------|-----------|--------------------|--------------------|--------------------|--------------------|
| F26      | NOGWO     | $4.86 \times 10^3$ | $6.08 \times 10^2$ | $6.19 \times 10^3$ | $3.60 \times 10^3$ |
|          | GWO       | $4.94 \times 10^3$ | $6.39 \times 10^2$ | $6.41 \times 10^3$ | $3.43 \times 10^3$ |
|          | DE        | $5.56 \times 10^3$ | $1.07 \times 10^2$ | $5.81 \times 10^3$ | $5.32 \times 10^3$ |
|          | WOA       | $5.69 \times 10^3$ | $9.53 \times 10^2$ | $7.39 \times 10^3$ | $3.55 \times 10^3$ |
|          | ALO       | $5.87 \times 10^3$ | $8.93 \times 10^2$ | $7.16 \times 10^3$ | $2.80 \times 10^3$ |
|          | SCA       | $7.84 \times 10^3$ | $4.43 \times 10^2$ | $8.56 \times 10^3$ | $6.63 \times 10^3$ |
| F27      | NOGWO     | $3.26 \times 10^3$ | $4.05 \times 10^1$ | $3.41 \times 10^3$ | $3.22 \times 10^3$ |
|          | GWO       | $3.27 \times 10^3$ | $2.92 \times 10^1$ | $3.36 \times 10^3$ | $3.23 \times 10^3$ |
|          | DE        | $3.23 \times 10^3$ | $6.70 \times 10^0$ | $3.24 \times 10^3$ | $3.22 \times 10^3$ |
|          | WOA       | $3.29 \times 10^3$ | $3.77 \times 10^1$ | $3.38 \times 10^3$ | $3.24 \times 10^3$ |
|          | ALO       | $3.42 \times 10^3$ | $9.11 \times 10^1$ | $3.73 \times 10^3$ | $3.25 \times 10^3$ |
|          | SCA       | $3.53 \times 10^3$ | $6.78 \times 10^1$ | $3.71 \times 10^3$ | $3.40 \times 10^3$ |
| F28      | NOGWO     | $3.45 \times 10^3$ | $6.13 \times 10^1$ | $3.64 \times 10^3$ | $3.33 \times 10^3$ |
|          | GWO       | $3.46 \times 10^3$ | $7.57 \times 10^1$ | $3.61 \times 10^3$ | $3.33 \times 10^3$ |
|          | DE        | $3.30 \times 10^3$ | $2.31 \times 10^1$ | $3.35 \times 10^3$ | $3.27 \times 10^3$ |
|          | WOA       | $3.45 \times 10^3$ | $7.60 \times 10^1$ | $3.64 \times 10^3$ | $3.34 \times 10^3$ |
|          | ALO       | $3.36 \times 10^3$ | $5.62 \times 10^1$ | $3.52 \times 10^3$ | $3.25 \times 10^3$ |
|          | SCA       | $4.43 \times 10^3$ | $4.14 \times 10^2$ | $5.67 \times 10^3$ | $3.68 \times 10^3$ |
| F29      | NOGWO     | $3.92 \times 10^3$ | $2.29 \times 10^2$ | $4.54 \times 10^3$ | $3.56 \times 10^3$ |
|          | GWO       | $4.01 \times 10^3$ | $2.27 \times 10^2$ | $4.61 \times 10^3$ | $3.61 \times 10^3$ |
|          | DE        | $4.16 \times 10^3$ | $1.23 \times 10^2$ | $4.50 \times 10^3$ | $3.91 \times 10^3$ |
|          | WOA       | $4.49 \times 10^3$ | $3.85 \times 10^2$ | $5.50 \times 10^3$ | $3.74 \times 10^3$ |
|          | ALO       | $4.72 \times 10^3$ | $4.11 \times 10^2$ | $5.65 \times 10^3$ | $4.14 \times 10^3$ |
|          | SCA       | $5.18 \times 10^3$ | $3.79 \times 10^2$ | $5.86 \times 10^3$ | $4.44 \times 10^3$ |
| F30      | NOGWO     | $1.31 \times 10^7$ | $1.34 \times 10^7$ | $5.61 \times 10^7$ | $8.25 \times 10^5$ |
|          | GWO       | $1.40 \times 10^7$ | $1.27 \times 10^7$ | $5.50 \times 10^7$ | $6.21 \times 10^5$ |
|          | DE        | $4.53 \times 10^5$ | $3.15 \times 10^5$ | $1.37 \times 10^6$ | $1.07 \times 10^5$ |
|          | WOA       | $4.24 \times 10^6$ | $2.91 \times 10^6$ | $1.04 \times 10^7$ | $4.27 \times 10^5$ |
|          | ALO       | $6.87 \times 10^6$ | $4.96 \times 10^6$ | $2.48 \times 10^7$ | $1.03 \times 10^6$ |
|          | SCA       | $1.97 \times 10^8$ | $7.07 \times 10^7$ | $3.80 \times 10^8$ | $9.68 \times 10^7$ |

Table 4. The Data analysis results (w/t/l).

| Algorithm | Mean    | Std    | Max     | Min     |
|-----------|---------|--------|---------|---------|
| NOGWO     | 10/1/19 | 4/0/26 | 4/0/26  | 12/0/18 |
| GWO       | 6/1/23  | 0/0/30 | 4/0/26  | 7/0/23  |
| DE        | 7/0/23  | 24/0/6 | 18/0/12 | 3/0/27  |
| WOA       | 1/0/29  | 0/0/30 | 1/0/29  | 3/0/27  |
| ALO       | 5/0/25  | 2/0/28 | 3/0/27  | 5/0/25  |
| SCA       | 0/0/30  | 0/0/30 | 0/0/30  | 0/0/30  |

According to Tables 3 and 4 and Figure 3, compared with some advanced swarm intelligence algorithms, NOGWO still performs very well.

For the average value, it is the most direct assessment indicator that illustrates the algorithm's exploration capacity. Among the experimental outcomes conducted by NOGWO, GWO, WOA, DE, ALO and SCA, NOGWO has the best presentation in average value, obtaining the best value on 10 test functions: F3, F11, F14, F17, F20, F18, F21, F29, F26 and F22. The NOGWO's outstanding presentation in average value fully indicates that the opposition-based learning model strategy influenced by refraction principle has obtained great outcomes on NOGWO, after incorporating an opposition-based learning model strategy influenced by refraction principle, the NOGWO's population multiplicity increases, the global exploration capacity is significantly better than GWO, and it is easier to leap out of local finest solution and find better solutions.



**Figure 3.** The comparison of NOGWO and other excellent algorithms.

For the minimum value, it can illustrate the algorithm's capacity to explore the global optima. NOGWO's presentation is still very outstanding in minimum value, in the comparison with GWO, WOA, DE, ALO and SCA. NOGWO obtains the minimum value on 12 test functions: F9, F3, F29, F21, F27, F22, F24, F18, F23, F11, F17, F13. Among these 12 test functions, 10 Functions are Hybrid Functions or Composition Functions. It implies that in complicated situations, NOGWO frequently exhibits enhanced global exploration capacity.

For standard deviation, standard deviation is utilized to assess the algorithm's robustness. The above experimental outcomes reveal that NOGWO's standard deviation performance is not as outstanding as the minimum and average values. While NOGWO's standard deviation outperforms GWO, it still trails behind DE.

In summary, compared with other outstanding swarm intelligence algorithms, NOGWO performs very well in global exploration, with stronger optimization ability and better convergence precision.

#### 4.3.1. Convergence Analysis

To analyze NOGWO's convergence, NOGWO and five other swarm intelligence algorithms were run on 30 CEC2017 test functions; Figure 4 and Appendix A display the convergence graphs. As shown in Figure A1 and Appendix A, NOGWO performs very well concerning convergence precision. On most test functions, NOGWO has the best convergence precision amidst the six swarm intelligence algorithms. On some test functions (such as F5, F10, F14, F16, F20, F24), in iteration's final phase, when other algorithms settle into the local finest solution and stop converging, NOGWO can successfully leap out of the local finest solution and continue to converge. Therefore, it means the opposition-based learning model strategy influenced by refraction principle has achieved good outcomes on NOGWO. It fully illustrates that NOGWO has outstanding global exploration capacity.

Moreover, NOGWO also performs well in terms of convergence speed. In testing functions F7, F11, F12, F21 and F28, the NOGWO's convergence curve gradient during the initial iteration is significantly higher than other algorithms, indicating that the NOGWO's convergence speed is rapider than other algorithms. However, on a few test functions (such as F1, F25, F28), NOGWO has undesirable convergence speed and precision, settling into the local finest solution and leading to early convergence.

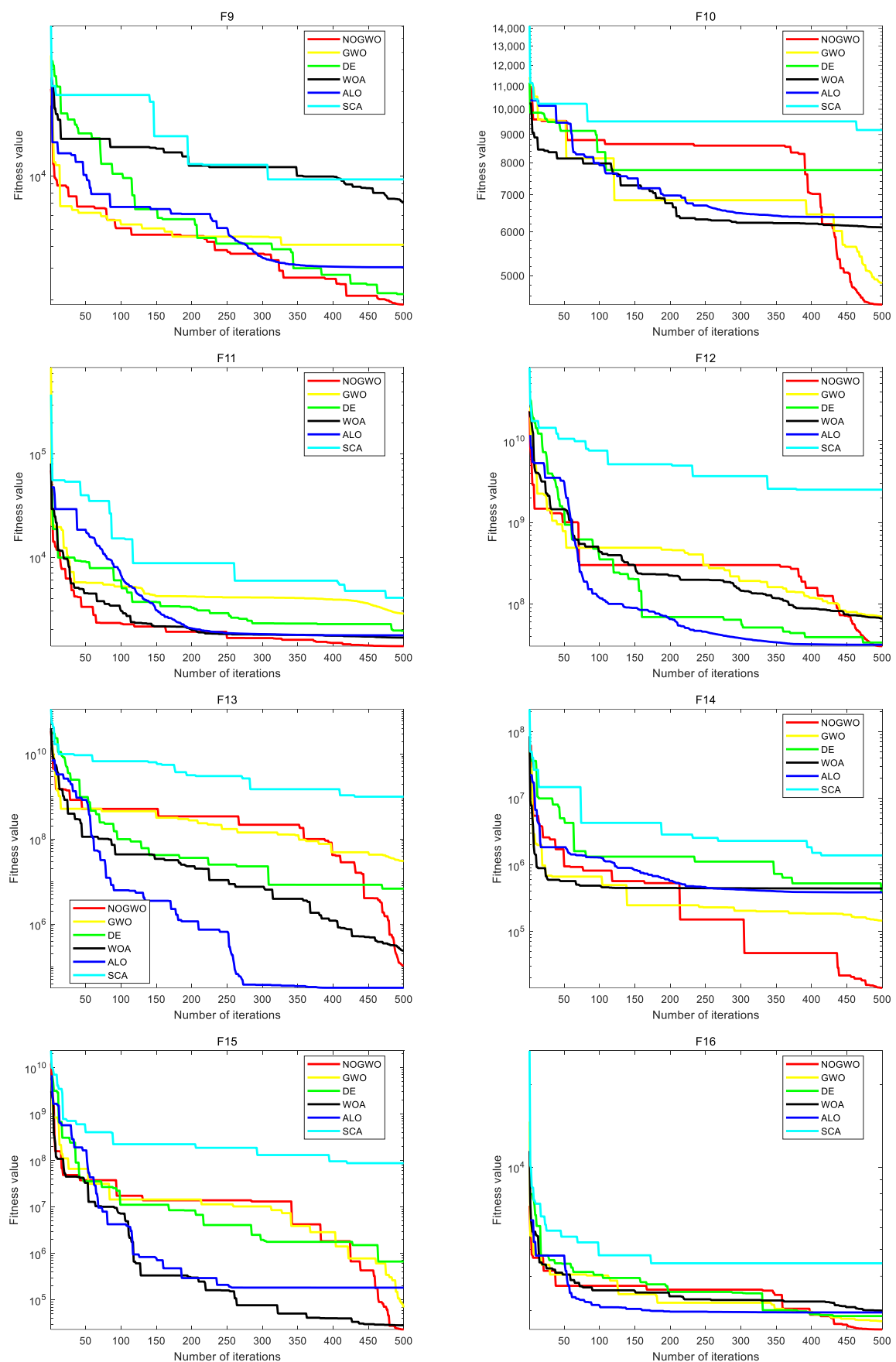


Figure 4. Convergence diagram.

#### 4.3.2. Friedman Test

To fully verify experimental outcomes and prove that the excellent effect achieved by NOGWO is not accidental, this article utilizes the Friedman test [22] for verification. The Friedman test was proposed by statistician Milton Friedman in 1937 to compare the differences between three or more samples. The mean rank is an important indicator for measuring sample diversity in the Friedman test. The less the mean rank of the sample, the superior the algorithm is, indicating that it is different from others. Under the premise of a confidence interval of 95%, Table 5 presents the six swarm intelligent algorithms' Friedman test outcomes.

**Table 5.** The results of the Friedman test.

| Algorithm | The Mean of Ranks |
|-----------|-------------------|
| NOGWO     | 2.40              |
| GWO       | 2.97              |
| DE        | 2.83              |
| WOA       | 3.70              |
| ALO       | 3.20              |
| SCA       | 5.90              |

According to Table 5, it can be seen that among these six swarm intelligence algorithms, NOGWO has the smallest average rank value. Therefore, it can be proven that compared with GWO, DE, WOA, ALO and SCA, NOGWO has significant differences and is the best.

#### 4.3.3. Engineering Application

At present, swarm intelligence algorithms are extensively utilized to settle various engineering optimization tasks. In this paper, to assess NOGWO's ability in practical applications, three engineering optimization problems are introduced for testing. Among them, the population scale is 50 and the max iteration count is 500. The test outcomes indicate that NOGWO has good practicality.

##### (1) Welding beam design

The welding beam design aims to create the welding beam at the most cost-effective price. This problem includes four independent variables  $l_1$ ,  $l_2$ ,  $l_3$ ,  $l_4$  and seven constraint conditions  $o_1$ ,  $o_2$ ,  $o_3$ ,  $o_4$ ,  $o_5$ ,  $o_6$ ,  $o_7$ . The problem's depiction is shown in Equation (16).

$$\begin{aligned}
 \text{cost} \quad & F(l) = 1.1047l_1^2l_2 + 0.04811l_3l_4(14.0 + l_2) \\
 \text{s.t} \quad & o_1(l) = \tau(l) - 13600 \leq 0, \\
 & o_2(l) = \sigma(l) - 30000 \leq 0, \\
 & o_3(l) = \delta(l) - 0.25 \leq 0, \\
 & o_4 = l_1 - l_4 \leq 0, \\
 & o_5(l) = P - P_c \leq 0, \\
 & o_6(l) = 0.125 - l_1 \leq 0, \\
 & o_7(l) = 1.1047l_1^2 + 0.04811l_3l_4(14.0 + l_2) - 5.0 \leq 0, \\
 \text{rang} \quad & 0.1 \leq l_1 \leq 2.0, \\
 & 0.1 \leq l_2 \leq 10.0, \\
 & 0.1 \leq l_3 \leq 10.0, \\
 & 0.1 \leq l_4 \leq 2.0.
 \end{aligned} \tag{16}$$

Table 6 presents welding beam design problem's outcomes. From Table 6, it is visible that compared with the other five swarm intelligence algorithms, NOGWO can design the welding beam with the lowest cost.

**Table 6.** Welded beam design problem.

| Algorithm | $l_1$  | $l_2$  | $l_3$   | $l_4$  | Optimal Cost   |
|-----------|--------|--------|---------|--------|----------------|
| NOGWO     | 0.2055 | 3.2398 | 9.0365  | 0.2057 | <b>1.69320</b> |
| GWO       | 0.2042 | 3.2687 | 9.0347  | 0.2059 | 1.69615        |
| DE        | 0.2093 | 2.9788 | 9.5114  | 0.2127 | 1.79684        |
| WOA       | 0.2041 | 3.2696 | 9.0366  | 0.2057 | 1.69513        |
| ALO       | 0.1342 | 5.8565 | 9.0367  | 0.2057 | 1.89244        |
| SCA       | 0.1645 | 4.0262 | 10.0000 | 0.2036 | 1.88628        |

**(2) Pressure vessel design**

Pressure vessel design's target is to descend production expense of pressure vessels. This task includes four independent variants  $g_1$ ,  $g_2$ ,  $g_3$ ,  $g_4$  and four constraint conditions  $c_1$ ,  $c_2$ ,  $c_3$ ,  $c_4$ . The description of this problem is shown in Equation (17).

$$\begin{aligned}
 \text{cost } & F(g) = 0.6224g_1g_3g_4 + 1.7781g_2g_3^3 + 3.1661g_1^2g_4 + 19.84g_1^2g_3 \\
 \text{s.t } & c_1(g) = -g_1 + 0.0193g_3 \leq 0, \\
 & c_2(g) = -g_3 + 0.00954g_3 \leq 0, \\
 & c_3(g) = -\Pi g_3^2g_4 - \frac{4}{3}\Pi g_3^3 + 1296000 \leq 0, \\
 & c_4(g) = g_4 - 240 \leq 0, \\
 \text{rang } & 0 \leq g_1 \leq 99, \\
 & 0 \leq g_2 \leq 99, \\
 & 10 \leq g_3 \leq 200, \\
 & 10 \leq g_4 \leq 200.
 \end{aligned} \tag{17}$$

Table 7 displays the outcomes of the pressure vessel design issue. Table 7's information indicates that NOGWO can significantly reduce the cost of pressure vessel design and has stronger practicality.

**Table 7.** Pressure vessel design problem.

| Algorithm | $g_1$    | $g_2$    | $g_3$     | $g_4$      | Optimal Cost     |
|-----------|----------|----------|-----------|------------|------------------|
| NOGWO     | 0.798244 | 0.395343 | 41.348645 | 186.193959 | <b>5925.2240</b> |
| GWO       | 1.012374 | 0.501105 | 52.437737 | 80.189589  | 6426.0690        |
| DE        | 0.813236 | 0.402309 | 42.067616 | 177.376991 | 5966.1910        |
| WOA       | 1.107443 | 0.547525 | 57.378749 | 48.797571  | 6720.8183        |
| ALO       | 1.064398 | 0.526132 | 55.149986 | 62.099404  | 6576.6354        |
| SCA       | 1.183176 | 0.539466 | 53.482687 | 74.356738  | 7487.3155        |

**(3) Design of compression/tension spring**

The object of the compression/tension spring task is to discover a tension/compression spring design with the lowest cost while meeting the corresponding constraint conditions. The description of this problem is shown in Equation (18).

$$\begin{aligned}
 \text{cost } & F(l) = (l_3 + 2)l_2l_1^2 \\
 \text{s.t } & c_1(l) = 1 - \frac{l_2^3l_3}{71785l_1^4} \leq 0, \\
 & c_2(l) = \frac{4l_2^2 - l_1l_2}{12566(l_2l_3^3 - l_1^4)} + \frac{1}{510l_1^2} \leq 0, \\
 & c_3(l) = 1 - \frac{140.45l_1}{l_2^2l_3} \leq 0, \\
 & c_4(l) = \frac{l_1 + l_2}{1.5} - 1 \leq 0, \\
 \text{rang } & 0.05 \leq l_1 \leq 2.00, \\
 & 0.25 \leq l_2 \leq 1.30, \\
 & 2.00 \leq l_3 \leq 15.0.
 \end{aligned} \tag{18}$$

The data in Table 8 indicates that in the tension/compression spring task, NOGWO has found the design that minimizes the weight of the tension/compression springs.

**Table 8.** Compression/tension spring design problem.

| Algorithm | $l_1$    | $l_2$    | $l_3$     | Optimal Cost     |
|-----------|----------|----------|-----------|------------------|
| NOGWO     | 0.051955 | 0.363144 | 10.923999 | <b>0.0126688</b> |
| GWO       | 0.050000 | 0.317296 | 14.047554 | 0.0127296        |
| DE        | 0.056979 | 0.497766 | 6.151031  | 0.0131727        |
| WOA       | 0.054967 | 0.440819 | 7.649949  | 0.0128525        |
| ALO       | 0.055722 | 0.461732 | 7.030393  | 0.0129465        |
| SCA       | 0.050000 | 0.314130 | 14.559287 | 0.0130044        |

## 5. Application of Path Planning for Unmanned Aerial Vehicles

Lately, unmanned aerial vehicles technology has developed rapidly and is extensively utilized in agriculture, military, intelligent manufacturing and other domains. The route planning of unmanned aerial vehicles is essential in the flight of unmanned aerial vehicles. Excellent flight paths not only ensure the safety of unmanned aerial vehicles, but also reduce fuel consumption, shorten flight time and improve the efficiency of unmanned aerial vehicle operation. Unmanned aerial vehicle path planning has become a hot topic in the application of swarm intelligence algorithms.

### 5.1. Description of UAV Path Planning Constraints

#### 5.1.1. Path Length Constraint

In unmanned aerial vehicle path planning, the flight path  $X_i$  is divided into  $n$  route points. The position of each route point is represented by a three-dimensional coordinates  $P_{ij} = (x_{ij}, y_{ij}, z_{ij})$ . The Euclidean distance between adjacent route points is the length of each flight path segment, so the length constraint is essentially the lengths of  $n - 1$  flight segments' aggregate. The equation for computing the length constraint of path  $X_i$  states in Equation (19).

$$\text{Cos } t_1(X_i) = \sum_{j=1}^{n-1} \left\| \overrightarrow{R_{ij}R_{i,j+1}} \right\| \quad (19)$$

#### 5.1.2. Threat Constraint

In unmanned aerial vehicle's flying, various obstacles are often encountered. In this paper, the set  $T = \{T_1, T_2, T_3, \dots, T_K\}$  is utilized to represent  $k$  obstacles, all of which are set as cylinders with a radius of  $L_k$  and a projection center coordinate point of  $C_k$ . Obstacles pose a threat to the unmanned aerial vehicle's security. Therefore, to ensure unmanned aerial vehicle's security, threat constraints must be set. The calculation formula for the threat constraint of the path  $X_i$  is shown in Equation (20).

$$\left\{ \begin{array}{l} \text{Cos } t_2(X_i) = \sum_{j=1}^{n-1} \sum_{k=1}^K T_k(\overrightarrow{R_{ij}R_{i,j+1}}) \\ T_k(\overrightarrow{R_{ij}R_{i,j+1}}) = \begin{cases} 0, d_k \geq S + D + L_k \\ (S + D + L_k) - d_k, D + L_k < d_k \leq S + D + L_k \\ \infty, d_k \leq D + L_k \end{cases} \end{array} \right. \quad (20)$$

where  $D$  is the unmanned aerial vehicle's diameter,  $S$  is a safe distance,  $\overrightarrow{R_{ij}R_{i,j+1}}$  is the  $j$ th path segment of the path  $X_i$  and  $d_k$  represents the length between the midpoint of  $\overrightarrow{R_{ij}R_{i,j+1}}$  and the  $k$ th threat center  $C_k$ .

### 5.1.3. Height Constraint

In the flight of unmanned aerial vehicles, if the flying altitude is too high, it is easy to be detected by the enemy. If the flying altitude is overly low, it is likely to crash into obstacles. Consequently, to ensure the safety of unmanned aerial vehicles, the flying elevation is typically constrained within a specific range. The equation for computing the height constraint of path  $X_i$  states in Equation (21).

$$\begin{cases} \cos t_3(X_i) = \sum_{j=1}^n H_{ij} \\ H_{ij} = \begin{cases} \left| h_{ij} - \frac{h_{\min} + h_{\max}}{2} \right|, & h_{\min} < h_{ij} < h_{\max} \\ \infty, & \text{else} \end{cases} \end{cases} \quad (21)$$

where  $h_{ij}$  is the unmanned aerial vehicle's flying altitude at the  $j$ th route point. In this section, the minimum altitude for unmanned aerial vehicle flight is  $h_{\min}$  and the maximum altitude is  $h_{\max}$ .

### 5.1.4. Mobility Constraint

In actual flight, due to the influence of obstacles, the flight path of unmanned aerial vehicles is often tortuous. Therefore, turning and climbing angles is critical in unmanned aerial vehicle's route planning. The mobility constraint of unmanned aerial vehicle flight mainly depends on turning and climbing angles. In this section, turn angle  $\alpha$  is the angle between two adjacent flight route segments' projections on the Oxy coordinate plane. The calculation formula is shown in Equation (22).

$$\alpha_{ij} = \arctan \left( \frac{\left\| \overrightarrow{R'_{ij}R'_{i,j+1}} \times \overrightarrow{R'_{i,j+1}R'_{i,j+2}} \right\|}{\left\| \overrightarrow{R'_{ij}R'_{i,j+1}} \cdot \overrightarrow{R'_{i,j+1}R'_{i,j+2}} \right\|} \right) \quad (22)$$

where  $\alpha_{ij}$  is the turning angle of the  $j$ th route point in flight path  $X_i$ ,  $\overrightarrow{R'_{ij}R'_{i,j+1}}$  and  $\overrightarrow{R'_{i,j+1}R'_{i,j+2}}$  are the projections of the flight path segments  $\overrightarrow{R_{ij}R_{i,j+1}}$  and  $\overrightarrow{R_{i,j+1}R_{i,j+2}}$  on the Oxy coordinate plane, respectively.

The mathematical expression for the climbing angle  $\beta$  is shown in Equation (23).

$$\beta_{ij} = \arctan \left( \frac{z_{i,j+1} - z_{ij}}{\left\| \overrightarrow{R'_{ij}R'_{i,j+1}} \right\|} \right) \quad (23)$$

where  $z_{i,j+1}$  and  $z_{ij}$  typify the z-axis coordinates of the  $(j+1)$ th and  $j$ th route points in the flight path  $X_i$ , respectively.

The calculation formula for the mobility constraint of the path  $X_i$  is shown in Equation (24).

$$\cos t_4(X_i) = a_1 \sum_{j=1}^{n-2} \alpha_{ij} + a_2 \sum_{j=1}^{n-1} |\beta_{ij} - \beta_{i,j-1}| \quad (24)$$

where  $a_1$  and  $a_2$  are coefficients of turning angle and climbing angle, respectively.



### 5.1.5. Total Cost Function

The total cost function is utilized to quantify the path planning task. The lower the cost function's value is, the superior the flight path is. We use NOGWO and other excellent algorithms to optimize the cost function with the aim of obtaining the optimal UAV flight path. The formula for the total cost function is shown in Equation (25).

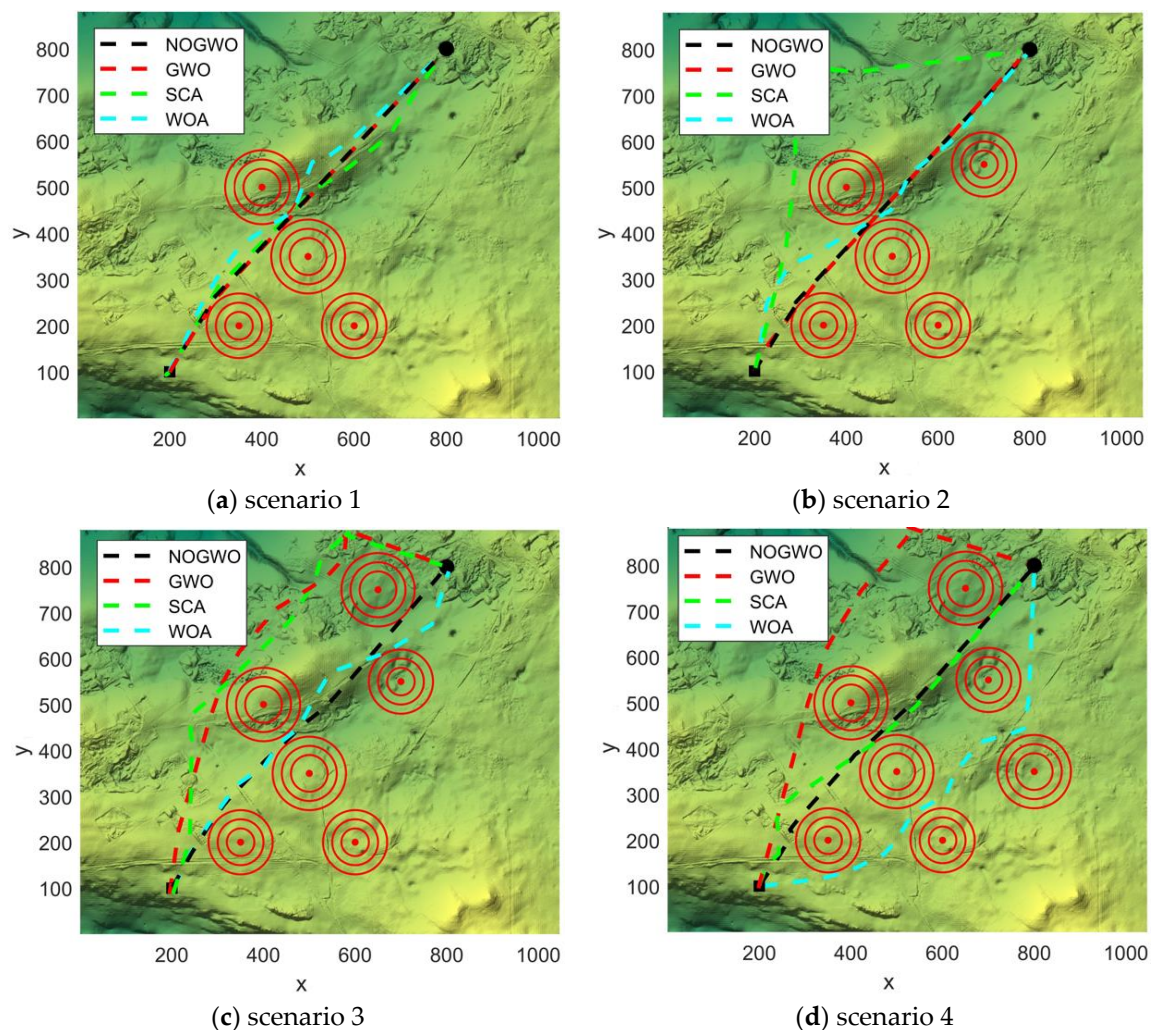
$$\text{Cost}(X_i) = c_1 \text{Cost}_1(X_i) + c_2 \text{Cost}_2(X_i) + c_3 \text{Cost}_3(X_i) + c_4 \text{Cost}_4(X_i) \quad (25)$$

where  $c_1, c_2, c_3$  and  $c_4$  are all weight coefficients.  $X_i$  represents the flight path.  $\text{Cost}_1(X_i)$ ,  $\text{Cost}_2(X_i)$ ,  $\text{Cost}_3(X_i)$ ,  $\text{Cost}_4(X_i)$  represents the length constraint cost, threat constraint cost, altitude constraint cost and mobility constraint cost of the flight path  $X_i$ , respectively.

## 5.2. Simulation Experiments

### 5.2.1. Scenario Setup and Parameter Settings

In this section, the scenarios utilized to conduct experiments are built upon real digital elevation model (DEM) maps that come from LiDAR sensors [23]. A part of an island is selected for modeling to obtain the actual scenario model. Then, by setting cylindrical obstacles to expand the scenario, four different scenarios are obtained, as shown in Figure 5. Scenarios 1, 2, 3 and 4 have 4, 5, 6 and 7 cylindrical obstacles respectively. In Figure 5, each red circle represents a cylindrical obstacle, the origin and destination of the flight route are represented by black solid squares and black solid circles, respectively.



**Figure 5.** Top view of paths generated by NOGWO and other swarm intelligence algorithms in scenarios 1 to 4.



To better test the practicality of NOGWO in unmanned aerial vehicle path planning problems, this section introduces three swarm intelligent algorithms: GWO, SCA and WOA, to compare with NOGWO. The population scale is 500, the iteration count is 200, and the flight path segments are 10 and 20, respectively. Repeat each algorithm 10 times, record the minimum path cost obtained from each run and calculate their average, standard deviation, maximum and minimum values.

### 5.2.2. Simulation Experiment Results and Analysis

Table 9 displays the statistical outcomes of the flight path cost of the four swarm intelligent algorithms in different scenarios with a flight path segment of 10. By analyzing the table data, it can be found that in all four scenarios, the average cost of flight paths generated using NOGWO is the smallest. Moreover, the standard deviation of flight path cost generated using NOGWO is also the smallest, indicating that NOGWO not only performs well in unmanned aerial vehicle path planning but is also very stable. It is worth noting that in more complex scenarios, such as scenario 3 and scenario 4, NOGWO outperforms the other three swarm intelligence algorithms. It means that in complex scenarios, NOGWO can often find better flight paths in unmanned aerial vehicle route planning.

**Table 9.** The outcomes of the simulation experiment when the route segment is 10.

| Scenario | Algorithm | Mean           | Std           | Max            | Min            |
|----------|-----------|----------------|---------------|----------------|----------------|
| 1        | NOGWO     | <b>4656.67</b> | <b>1.55</b>   | <b>4659.23</b> | <b>4655.22</b> |
|          | GWO       | 4658.20        | 1.71          | 4661.01        | 4656.00        |
|          | SCA       | 5685.49        | 502.34        | 6356.14        | 4786.08        |
|          | WOA       | 8391.67        | 642.41        | 9779.49        | 7526.52        |
| 2        | NOGWO     | <b>4658.64</b> | <b>1.94</b>   | <b>4661.23</b> | <b>4656.01</b> |
|          | GWO       | 4757.21        | 176.37        | 5090.76        | 4657.26        |
|          | SCA       | 5408.48        | 388.92        | 6229.09        | 4926.61        |
|          | WOA       | 9002.88        | 809.73        | 10,646.21      | 7778.30        |
| 3        | NOGWO     | <b>5152.31</b> | <b>622.67</b> | <b>6151.81</b> | <b>4688.59</b> |
|          | GWO       | 7451.77        | 2390.37       | 11,001.30      | 4705.14        |
|          | SCA       | 8165.97        | 1122.61       | 9209.77        | 5608.55        |
|          | WOA       | 9739.69        | 932.96        | 11,202.52      | 8128.42        |
| 4        | NOGWO     | <b>4942.72</b> | <b>321.24</b> | <b>5617.18</b> | 4721.42        |
|          | GWO       | 6931.27        | 2296.35       | 10,911.31      | 4707.30        |
|          | SCA       | 8164.69        | 1192.57       | 9934.99        | 6427.13        |
|          | WOA       | 10,186.34      | 938.96        | 11,755.63      | 9043.92        |

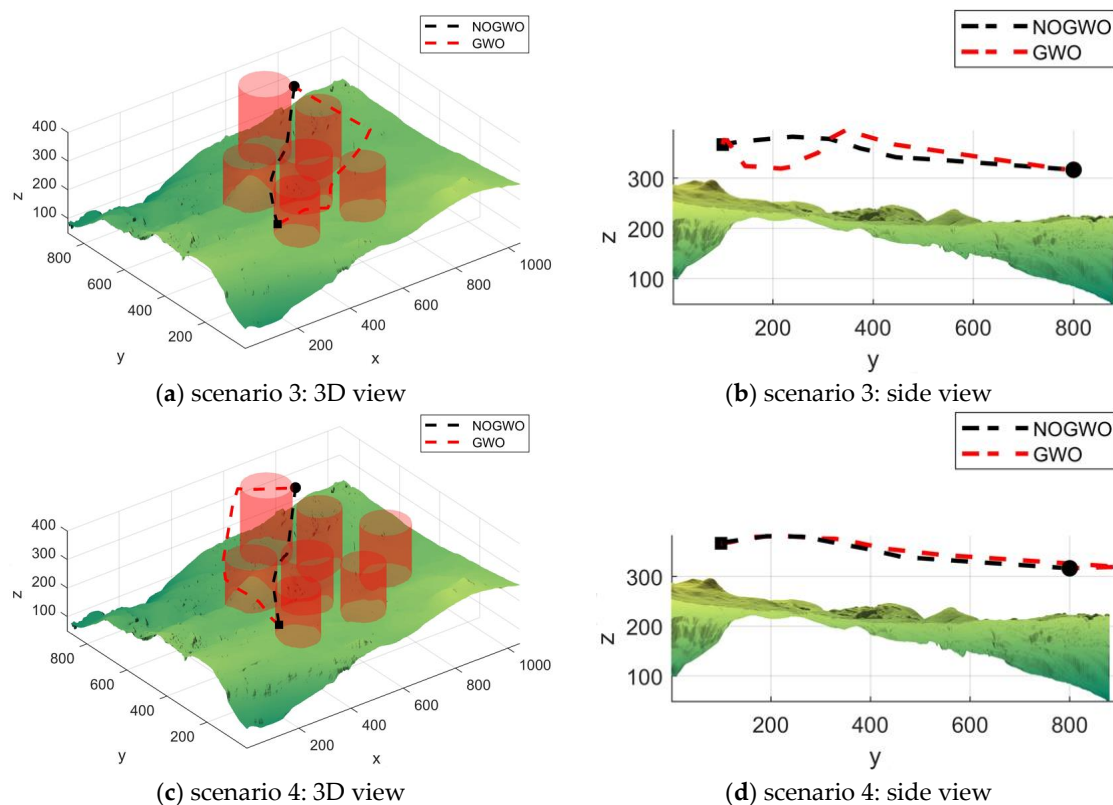
To demonstrate the scalability of NOGWO, the flight path segment is set to 20 for a deeper comparison. The statistical results of the flight path cost of the four swarm intelligent algorithms are shown in Table 10. According to Table 10, both WOA and SCA perform poorly in all four scenarios. For GWO, although there is not much difference in performance between GWO and NOGWO in simple scenarios such as scenario 1 and scenario 2, GWO's performance is far inferior to NOGWO when the scenarios are more complex, especially in scenario 3 and 4.

Therefore, in both simple and complex scenarios, NOGWO performs better in unmanned aerial vehicle path planning problems. It is mainly because after adopting the random walk strategy and the opposition-based learning strategy influenced by refraction principle, NOGWO has a larger exploration scope and greatly improves population multiplicity. Therefore, the likelihood of discovering the global optima is higher, and better unmanned aerial vehicle flight paths can be planned.

**Table 10.** The outcomes of the simulation experiment when the route segment is 20.

| Scenario | Algorithm | Mean           | Std            | Max            | Min            |
|----------|-----------|----------------|----------------|----------------|----------------|
| 1        | NOGWO     | <b>4714.33</b> | <b>139.28</b>  | <b>5108.24</b> | 4659.16        |
|          | GWO       | 4836.80        | 289.53         | 5392.45        | 4658.77        |
|          | SCA       | 8631.36        | 1523.72        | 11,221.17      | 7010.30        |
|          | WOA       | 11,934.01      | 1486.04        | 14,383.97      | 10,434.10      |
| 2        | NOGWO     | <b>4773.09</b> | <b>225.59</b>  | <b>5209.51</b> | <b>4657.75</b> |
|          | GWO       | 4824.12        | 256.22         | 5217.18        | 4659.03        |
|          | SCA       | 10,278.69      | 1250.53        | 11,898.81      | 7840.16        |
|          | WOA       | 12,369.43      | 1337.45        | 14,450.72      | 10,407.35      |
| 3        | NOGWO     | <b>5762.38</b> | <b>1011.61</b> | <b>6967.97</b> | <b>4714.00</b> |
|          | GWO       | 11,664.49      | 3335.20        | 14,650.18      | 5271.94        |
|          | SCA       | 11,966.29      | 1187.49        | 13,537.90      | 9743.38        |
|          | WOA       | 12,258.20      | 1219.40        | 14,635.69      | 9986.29        |
| 4        | NOGWO     | <b>6046.20</b> | 1417.89        | <b>9095.99</b> | <b>4714.91</b> |
|          | GWO       | 10,014.74      | 3523.07        | 14,066.93      | 5389.79        |
|          | SCA       | 11,788.33      | 1323.21        | 13,121.40      | 9235.97        |
|          | WOA       | 12,663.88      | 892.94         | 13,933.16      | 11,328.10      |

Figure 5 shows a top view of the four swarm intelligent algorithms generating flight paths in four different scenarios. From Figure 5, it can be seen that all four swarm intelligent algorithms can successfully avoid obstacles and generate flight routes from origin to destination. However, in most scenarios, the flight path generated by NOGWO is often the shortest and best. Figure 6 shows the side view and 3D image of the flight path generated by NOGWO in scenarios 3 and 4. It can be seen from Figure 6 that in more complex scenarios 3 and 4, the path generated by NOGWO can not only successfully avoid various obstacles, but also maintain a roughly equal distance from the undulating ground. Therefore, NOGWO has shown excellent performance in unmanned aerial vehicle path planning problems.

**Figure 6.** 3D view and side view of paths generated by NOGWO in scenarios 3 to 4.

## 6. Discussions

### 6.1. Experiment Outcomes

The outcomes in Tables 1–4 indicate that NOGWO has superior convergence accuracy and better global exploration capacity compared to GWO and its variants, as well as other excellent swarm intelligence algorithms.

Initially, NOGWO has superior convergence accuracy. Under traditional GWO principles, the alteration of the grey wolf's position is governed only by the three wolves with the best positions. As the top three wolves converge to a local optimum, the remaining wolves are swiftly drawn towards this local optimum due to the influence exerted by the top three wolves. In addressing this issue, the incorporation of an opposition-based learning model strategy influenced by refraction principle into NOGWO leads to the creation of a new solution for every existing solution, ultimately enriching the algorithm's population diversity. Hence, the grey wolf possesses a greater population size and a more extensive search region, thereby creating favorable conditions for identifying superior solutions. Overall, NOGWO demonstrates superior convergence accuracy compared to GWO and its iterations, a fact substantiated by the empirical findings presented in Tables 1 and 2.

Besides, NOGWO has better global exploration capacity. As depicted in Tables 3 and 4, the indicator minimum in experimental outcomes establishes that NOGWO achieves global optimum on 12 test functions and outperforms the 5 algorithms included in the comparison, thus providing evidence of the effectiveness of the random walk strategy in evading local optimum. By incorporating the random walk strategy, NOGWO introduces a more randomized update mechanism for the grey wolf's location, resulting in a substantial increase in the probability of discovering the global optima. Hence, NOGWO showcases remarkable global search proficiency and offers a greater prospect of identifying the global optima when compared to other algorithms. Furthermore, the nonlinear convergence factor strategy implemented in NOGWO effectively regulates the equilibrium between global exploration and local exploitation. In the early and middle phases of iteration, the nonlinear convergence factor maintains a proximity to 2, facilitating extensive exploration across the solution space and emphasizing global search. Subsequently, during the latter stages of iteration, the nonlinear convergence factor rapidly declines towards 0, promoting localized exploitation and efficient convergence towards the optimal solution.

Nonetheless, NOGWO possesses certain shortcomings. The outcomes depicted in Tables 3 and 4 indicate that in comparison to some outstanding swarm intelligence algorithms, NOGWO's stability is inadequate. This inadequacy stems from the fact that while the incorporation of the random walk strategy improves the algorithm's ability to avoid local optimum, it also introduces some irregular and unstable solutions to some extent, thereby impacting the algorithm's stability. Thus, there is a need for future enhancements to elevate NOGWO's stability.

### 6.2. Engineering Problems Application

The experimental outcomes of NOGWO in three engineering optimization problems indicate that NOGWO excels in all three engineering optimization tasks. It can be primarily attributed to the exceptional performance of NOGWO with respect to minimum value. Even with the fulfillment of multiple constraints, NOGWO can excel at determining the design solution with the minimal cost. By incorporating random walks and an opposition-based learning model strategy influenced by refraction principle, NOGWO can provide more design solutions, expand the search scope and facilitate attaining the optimal design solution in engineering tasks. It convincingly displays the effectiveness of the opposition-based learning model strategy influenced by refraction principle and the random walk strategy. In pressure vessel design, for example, the highly restricted search space imposed by four specific constraints makes finding the minimum-cost design highly challenging. Many swarm intelligence algorithms often fall into local optima prematurely due to this limited exploration space. However, with the support of a random walk strategy and opposition-based learning model strategy influenced by refraction principle, NOGWO typ-

ically escapes local optima and persists in searching for more optimal solutions. Therefore, NOGWO demonstrates remarkable effectiveness in pressure vessel design, achieving the optimal design.

### 6.3. UAV Path Planning Analysis

The outcomes of the simulation experiments for UAV path planning reveal that NOGWO can devise the optimal, safe and stable flight path for UAVs. Particularly in the intricate scenarios 3 and 4, NOGWO reduces flight path cost substantially compared to GWO, SCA and WOA. NOGWO's exceptional success in UAV path planning can be credited to two pivotal factors.

Firstly, numerous constraints, such as flight altitude, flight path length and turning angle, must be considered in the UAV path planning task. As the constraint conditions become more stringent, swarm intelligence algorithms frequently prematurely converge towards local optima while updating the population location, influenced by these constraints. In GWO, the algorithm's local search randomness is constrained in consequence of the population's location update being influenced only by the best three grey wolves. Hence, when confronted with numerous constraints, GWO struggles to escape the local optima. In contrast, the adoption of the random walk strategy in NOGWO significantly boosts the algorithm's potential to escape local optima and address the problem of premature convergence and stagnation while exploring flight paths. Furthermore, NOGWO has brought in a novel convergence factor that effectively regulates the exploitation and exploration capacities, extends the exploration scope and prevents premature convergence of the algorithm.

Secondly, within the realm of UAV path planning, numerous obstacles exist. As the obstacle's count escalates, the algorithm's exploration scope undergoes a marked shrinkage to maintain flight safety. Consequently, the distance among grey wolf population lessens, triggering heightened centralization of grey wolf population, diminished algorithm population diversity and elevated hazard of settling into local optima. Nevertheless, the incorporation of the opposition-based learning model strategy influenced by refraction principle in NOGWO yields an opposite solution for every individual population, markedly boosting the population's diversity, expanding exploration scope and presenting a greater variety of flight paths for selection, thus facilitating NOGWO in exploring the best flight path.

Nevertheless, this study also has some limitations. For example, this study does not consider the effect of wind [24] on path planning and trajectory. Unmanned aerial vehicles are highly vulnerable to wind disturbances, which can impact their attitude, velocity and position. Another critical aspect that this study overlooks is the geometry of generated obstacles [25]. Studies reveal that path-planning algorithms demonstrate improved performance in the presence of obstacles with lower levels of concavity. However, their performance is greatly diminished when confronted with complex, concave obstacles. Hence, this study's future direction is to consider the impact of wind and obstacle geometry on path planning.

## 7. Conclusions

To address grey wolf algorithm's drawbacks in complicated optimization tasks, such as settling into local optima easily and converging too early, this paper designs a novel algorithm, NOGWO. By introducing a nonlinear convergence factor strategy, an opposition-based learning model strategy influenced by refraction principle and a random walk strategy, a multi-strategy collaborative grey wolf optimization algorithm NOGWO is designed. Conducting experiments by utilizing 30 CEC2017 test functions, experimental outcomes indicate that NOGWO performs well concerning convergence precision and convergence swiftness. By conducting a Friedman test, it is verified that the excellent performance of NOGWO in the experiment was not accidental, and NOGWO outperforms the other five algorithms participating in comparison. Finally, apply NOGWO to three

engineering tasks and unmanned aerial vehicle path planning tasks, the outcomes show that NOGWO has good practicality.

However, NOGWO still has some shortcomings, as mentioned in the discussions and will continue to be improved and optimized in the future.

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**Conflicts of Interest:** The authors declare no conflict of interest.

## Appendix A

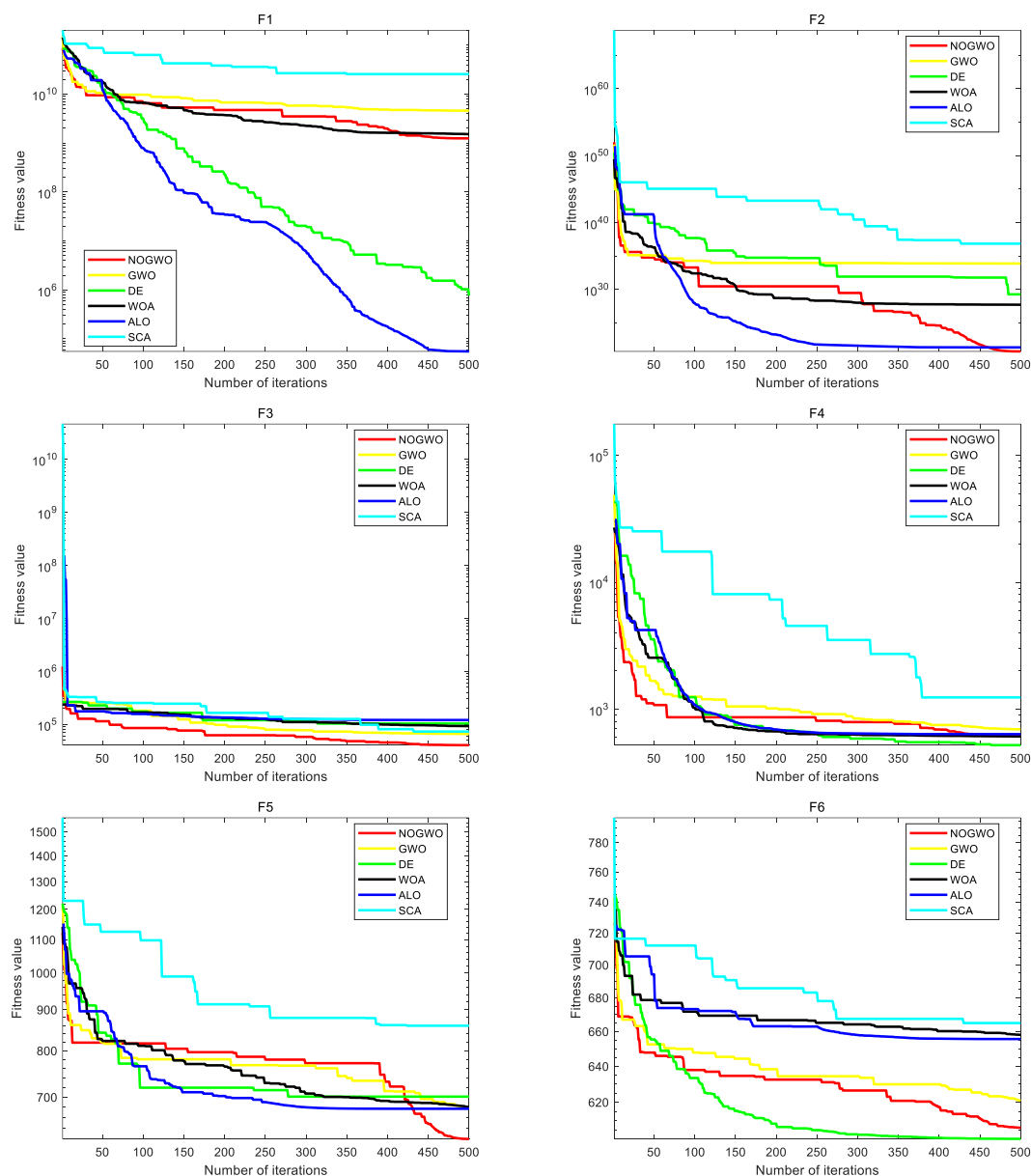


Figure A1. Cont.

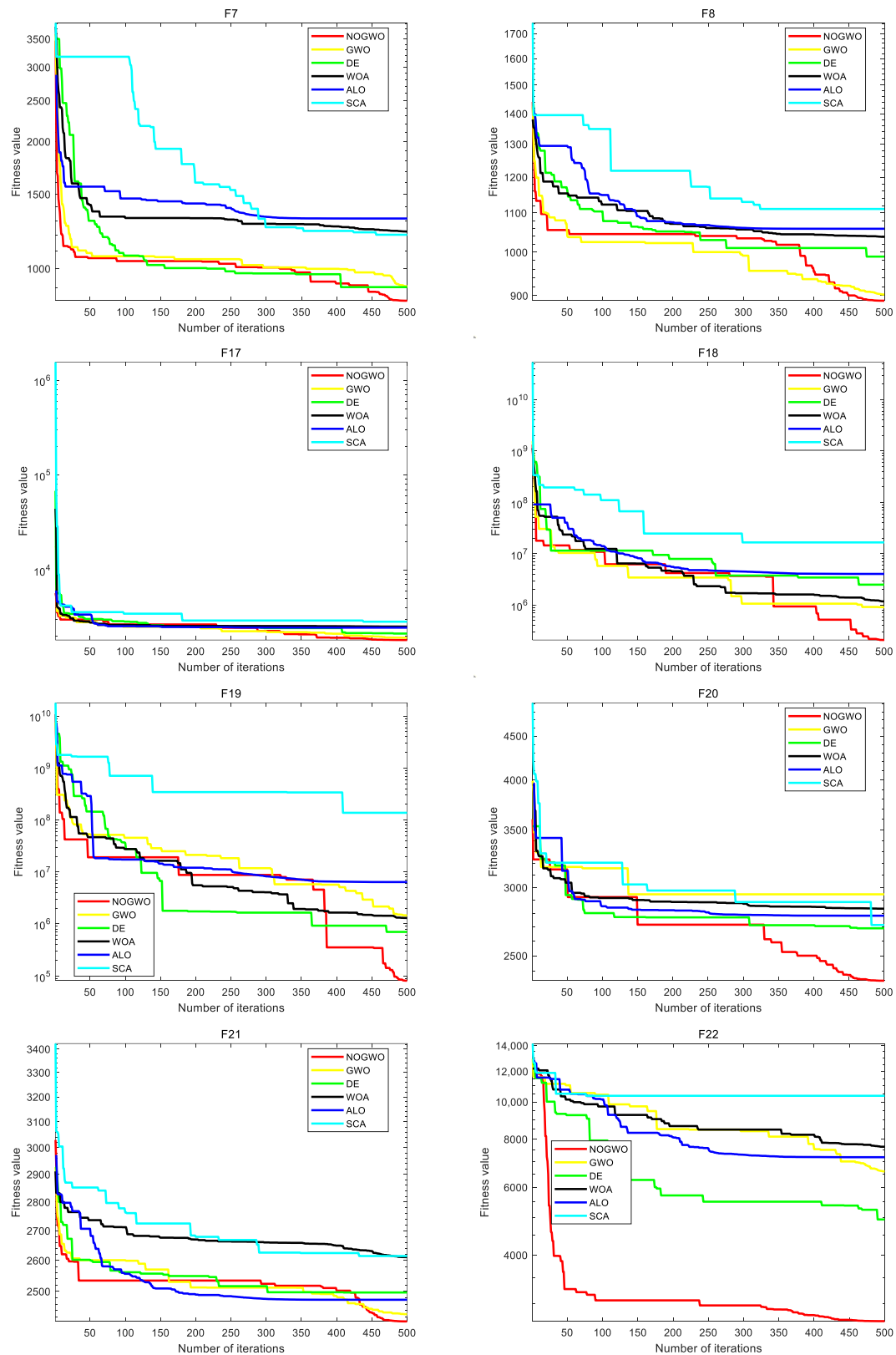


Figure A1. Cont.

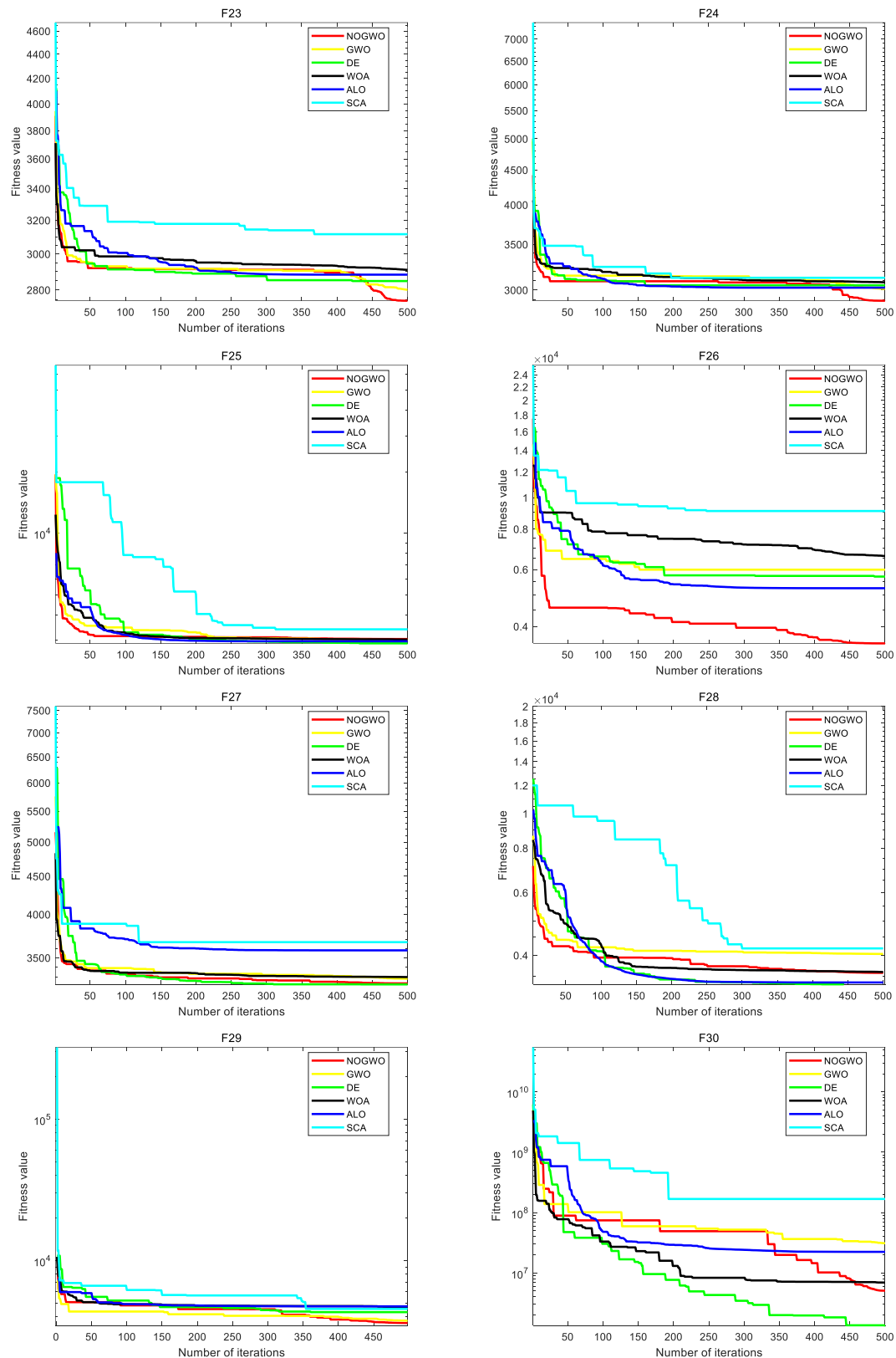


Figure A1. Convergence diagram.



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